

Soybean RIN4 represents a mechanistic link between plant immune and symbiotic signaling

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Katalin Tóth , Daewon Kim, Sung-Hwan Cho, Cuong T Nguyen, Tran HN Nguyen, Christopher Hartanto, Jean-Michel Michno, Adrian O Stec, Robert M Stupar, Gary Stacey 

Division of Plant Science and Technology, C.S. Bond Life Sciences Center, University of Missouri, Columbia, United States • Cuu Long Delta Rice Research Institute, Cần Thơ City, Vietnam • Department of Molecular and Environmental Biotechnology, University of Science, VNU HCMC, Ho Chi Minh City, Vietnam • Division of Biochemistry, University of Missouri, Columbia, United States • Department of Agronomy and Plant Genetics, University of Minnesota, St. Paul, United States

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The manuscript reports **fundamental** findings supported by **convincing** data that supports the biological mechanism for optimal nodulation in soybean. The results are of relevance to understanding the signaling pathways (specifically those dependent on RIN4/RPM1-interacting protein 4) underpinning beneficial rhizobia symbiosis, while repressing the immune response.

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Abstract

The legume-rhizobium symbiosis represents a unique and beneficial interaction between legumes and nitrogen-fixing soil bacteria, called rhizobia. The initiation and development of this symbiosis is complex and begins with recognition of key molecular signals, produced by the plant and its symbiont, which determine symbiotic compatibility. Current data suggest that the invading symbiont initially triggers plant immune responses that are subsequently suppressed. Hence, there is growing evidence that features of plant immunity may be relevant to symbiotic establishment. RIN4 is a key immune regulator in plants, regulating basal immunity and it is also targeted by pathogen effector proteins that either confer susceptibility or resistance, depending on the presence of the appropriate resistance protein. Surprisingly, we found that RIN4 was rapidly phosphorylated upon rhizobial inoculation of soybean root hairs. RNAi silencing and mutant studies indicate that RIN4 expression is essential for effective nodulation of soybean. RIN4 phosphorylation occurs within a fifteen amino acid motif, which is highly conserved within the Fabales (legumes) and Rosales orders, which comprise species capable of nitrogen-fixing endosymbiosis with rhizobia. RIN4 proteins mutated in this conserved phosphorylation site failed to support efficient soybean nodulation. Phosphorylation of this site is mediated by the symbiotic receptor-like kinase, SymRK, a well-studied member of the symbiotic signaling pathway. The data implicate RIN4

phosphorylation as a key mediator of rhizobial compatibility, interconnecting symbiotic and immune signaling pathways.

Significance

The nitrogen fixing legume-rhizobium symbiosis is a cornerstone of sustainable agriculture, with ongoing efforts to transfer this unique ability to non-leguminous crop plants. Plants are surrounded by a myriad of microbes in the soil, and, therefore, require constant surveillance in order to distinguish between a pathogen or symbiont. Plants monitor for specific molecular signals that indicate pathogen or symbiont presence. We show that RIN4, a key immune regulator, plays an essential role in promoting the development of the symbiotic nitrogen-fixing relationship between soybean and its compatible symbiont *Bradyrhizobium japonicum*. Therefore, RIN4 is likely a key player in mediating the appropriate response upon infection by friend or foe.

Introduction

At the very beginning of the symbiotic signaling pathway, there are three receptor-like kinases (RLKs) that are indispensable for the initiation of two developmental processes: bacterial infection and nodule organogenesis. Bacterial infection occurs primarily via an infection thread (IT) developed within the infected root hair. IT delivers rhizobia into the underlying newly divided cortical cells forming the nodule primordium, which occurs in parallel with bacterial infection (1; 2). Underneath the infection site, the nodule primordium develops into a new organ, the nodule, where the rhizobia are accommodated and convert atmospheric nitrogen into ammonia. As a result, a nitrogen-fixing symbiosis will develop (1; 2). In legumes with determinate nodules, such as soybean (*Glycine max*) and *Lotus japonicus*, two Lysin (LysM)-domain containing RLKs, Nodulation Factor Receptor 1 and 5 (NFR1 and NFR5) perceive the rhizobial lipo-chitooligosaccharide nodulation factor (NF) (3; 4; 5; 6). Rhizobia produce NF in response to flavonoids secreted by the host legume. Lotus or soybean mutants lacking NFR1 and/or NFR5 do not respond to rhizobial inoculation and do not form nodules (4). A third RLK, containing extracellular leucine-rich repeats, is located downstream of the NF receptors, called Symbiosis Receptor Kinase (SymRK) (7; 8). Root hairs of *Lotus symrk* mutant plants (i.e., *cac41.5* insertion mutant) do not curl and bacterial infection cannot occur despite displaying root hair deformation (7). *symrk-10*, a *Lotus* mutant carrying a point mutation in the activation loop of the kinase domain, abolishing kinase activity, displays a similar phenotype as the insertion mutant (9), underpinning the importance of phosphorylation in the symbiotic signaling cascade.

Studies using *Lotus* showed that SymRK interacts strongly with NFR5 and weakly with NFR1 (10). It was shown that autophosphorylation of NFR1 is essential for downstream signaling. NFR5 lacks kinase activity and is trans-phosphorylated by NFR1 (11), and by a third LysM-containing RLK, NFre (epidermal), enhancing the robustness of NF-signaling (12). Transducers of RLK-induced signaling are receptor-like cytoplasmic kinases (RLCKs), a NFR5-interacting cytoplasmic kinase 4 (NiCK4) was shown to be an important link between NF perception by NFR5 and nodule organogenesis (13). Briefly, downstream components of the pathway are a calcium- and calmodulin-dependent kinase, CCaMK which phosphorylates CYCLOPS, a DNA-binding transcriptional activator (14). The CCaMK/CYCLOPS complex controls bacterial infection as well as nodule organogenesis (14), forming a regulatory unit with other transcriptional regulators and activates NODULE INCEPTION (NIN), a nodulation specific transcription factor. NIN is involved in root hair and epidermal as well as cortical cell responses, the latter leading to nodule development (15).

Given the importance of phosphorylation in the symbiotic signaling cascade, it is not surprising that phosphoproteomic studies have been reported for a variety of legumes (16, 17, 18). The primary entry point for rhizobium in the case of *L. japonicus*, *Medicago truncatula*, and *G. max* is the root hair (19). However, only a small fraction of the root hairs on a given root are infected and even fewer infections lead to nodule formation (20). Hence, phosphoproteomic studies using entire roots, such as conducted with *L. japonicus* and *M. truncatula*, likely suffer from signal dilution due to the highly localized nature of rhizobial infection. Therefore, we previously performed phosphoproteomic studies of isolated soybean root hair cells (separated from the root), in order to reduce signal dilution due to non-responding root tissues (17). Indeed, this study identified a variety of proteins that were rapidly (one-hour post-inoculation) phosphorylated upon treatment with the compatible symbiont, *Bradyrhizobium japonicum*. To our surprise, among these proteins was the plant immune regulator RPM1-Interacting protein 4 (RIN4; 17).

RIN4 was discovered in *Arabidopsis thaliana* as an interactor of RPM1, a disease resistance protein conferring resistance against the bacterial leaf pathogen *Pseudomonas syringae* (21). RIN4 is conserved among land plants and is involved in the regulation of Pattern Triggered Immunity (PTI). Given RIN4's regulatory function in PTI, it is not surprising that the protein is targeted by several effector proteins released by pathogens to interfere and modulate plant immune responses (22). RIN4 undergoes post-translational modifications (PTM) or proteolytic cleavage as a consequence of being targeted by *P. syringae* effector proteins (23, 24). RIN4 modifications trigger a second layer of immune responses triggered by resistance (R) proteins, intracellular immune receptors (nucleotide-binding leucine-rich repeat receptor or NLRs) which monitor perturbations within the host cell leading to NLR-triggered immunity (NTI) (24). RIN4 is an intrinsically disordered protein (IDP) (25, 26). IDPs lack stable secondary and tertiary protein structure and can transition from disorder to order upon interactions with other protein(s) or upon PTMs like phosphorylation (27). Lee and colleagues (26) demonstrated that phosphorylated RIN4 is more flexible than native RIN4 contributing to its conformational flexibility and function.

P. syringae effector proteins AvrB and AvrRpm1 induce RIN4 phosphorylation, suppressing PTI responses (28, 29). Phosphorylation of AtRIN4 at serine 141 is triggered upon bacterial flagellin treatment (i.e., flagellin epitope, flg22). Phosphorylation of this site is the target of AvrB and consequently induces phosphorylation of the evolutionary conserved threonine 166. Increased T166 phosphorylation suppresses S141 induced PTI responses (29). It was shown that AvrRpm1 ADP-ribosylates *Arabidopsis* as well as soybean RIN4, and this ADP-ribosylation is a prerequisite for subsequent phosphorylation of the T166 phosphorylation site (30). AvrB-induced T166 phosphorylation is mediated by an RLCK, RIPK. However, in an *Arabidopsis ripk* mutant background, RIN4 phosphorylation in response to AvrB is decreased and not abolished (31). Furthermore, Xu and colleagues (32) showed that several other RLCKs were able to phosphorylate RIN4. Hence, the relative phosphorylation of S141 and T166 determines the plant response to pathogen effector proteins and subsequent disease progression.

Although RIN4 plays an essential role in pathogen virulence and host immunity, the details of RIN4 molecular function are not well understood (24). It should be noted that virtually all the studies on RIN4 have used leaves, the natural infection route for *P. syringae*.

The *A. thaliana* genome encodes a single *RIN4* gene. In contrast, the soybean genome encodes four *RIN4* (*GmRIN4a-d*) genes (33). In soybean, RPG1-B (resistance to *Pseudomonas syringae* pv *glycinea*) R protein conveys resistance to *P. syringae* expressing AvrB. Both GmRIN4a and GmRIN4b were shown to associate with AvrB, but only GmRIN4b interacts with RPG1-B (33). These data and the above mentioned AvrRpm1-mediated ADP-ribosylation of both At and GmRIN4 suggest that, at least in leaves, the GmRIN4 proteins play a role in plant immunity similar to that defined by detailed studies in *Arabidopsis*.

In the work described here, we demonstrate that soybean RIN4 (GmRIN4, hereafter RIN4) protein(s) are essential for efficient nodulation of soybean. This function of RIN4 is mediated by specific phosphorylation of serine 143, which is located within a 15 amino acid (aa) motif. These 15 aa are absent in *Arabidopsis*, and seem to be highly conserved within the Fabales and Rosales plant orders, therefore suggesting a symbiosis-related function. Phosphorylation of S143 is mediated by GmSymRK β . We found that *RIN4a* and *RIN4b* are highly expressed in root hair cells and that their expression level does not change upon rhizobial treatment. A soybean mutant line, in which *RIN4b* was mutated using CRISPR-Cas9 mediated gene editing, led to a significant reduction in nodulation concomitant with the reduction in the expression of downstream components of the symbiosis signaling pathway.

Results and Discussion

Soybean RIN4 proteins harbor a highly conserved and novel-RIN4-motif

A key, unifying feature of the nitrogen-fixing symbiosis is the formation of nodules where the bacteria are accommodated inside living plant cells (34). The symbiosis is restricted to one phylogenetic clade containing four orders: Fabales (legumes), Fagales, Cucurbitales and Rosales (FaFaCuRo). Within this clade, there are 10 families out of the 28 that contain species which form nitrogen-fixing root nodules (34). Legumes and the non-legume *Parasponia andersonii* (Rosales) form symbiosis with rhizobia. Actinobacteria, *Frankia*, interact with species from Rosales, Fagales and Cucurbitales (35), which also leads to intracellular symbiotic nitrogen fixation.

We built a phylogenetic tree (SI Appendix, Fig. S1 and Fig. 1A) of 149 RIN4 sequences derived from 66 species of the FaFaCuRo clade and species outside of the clade (Materials and Methods and SI Appendix, Table S1). The tree contains sequences from both nodulating and non-nodulating species of Fabales, Rosales, Fagales and Cucurbitales (35), as well as from species outside of the clade. This analysis identified a specific clade of RIN4 proteins (SI Appendix, Fig. S1). Two subclades were apparent: one of them comprised of sequences from Fabales (SI Appendix, Fig. S1, blue highlighted), the other contains sequences from Rosales (SI Appendix, Fig. S1, green highlighted). Within the sequences forming these two apparent subclades, we discovered a 15 amino acid motif, defined by a “GRDSP” core sequence (Fig. 1B, red box), suggesting a nodulation related function. We named it a novel-RIN4-motif (NRM) (Figure 1B, C). In Figure 1, the alignment shows RIN4 protein sequences from model legume species such as *G. max*, *P. vulgaris*, *L. japonicus* and *M. truncatula* aligned with *Arabidopsis* RIN4, and nodulating non-legume *P. andersonii* and its non-nodulating relative *Trema orientale* from Rosales. The sequence is absent in *Arabidopsis* RIN4 (Fig. 1B, red box) as well as in other non-FaFaCuRo species we used when building the phylogenetic tree. This motif is highly conserved among RIN4 proteins from nodulating and non-nodulating species of Fabales as well as Rosales. Interestingly, the motif is not conserved in RIN4 sequences derived from Fagales and Cucurbitales (SI Appendix, Fig S2). However, the motif is retained in non nodulating Fabales (such as *C. canadensis*, *N. schottii*; SI Appendix, Fig. S2) and Rosales species as well. NRM harbors the soybean RIN4 phosphorylation site, serine 143 (Fig. 1B, C) identified in our previous study of soybean root hairs (17). S143 is localized within the “GRDSP” core motif (Fig. 1B, red arrow), and is highly conserved across Fabales and Rosales.

RIN4a and *RIN4b* are highly expressed in soybean root hairs

There are four *RIN4* genes described in soybean (*RIN4a*, *RIN4b*, *RIN4c*, *RIN4d*; 33; Fig. 1). A characteristic feature of the AtRIN4 protein is two plant-specific nitrate-induced domains (NOI), an N-terminal and a C-terminal domain (Fig. 1B, grey underline; Fig 1C). AvrRpt2 bacterial

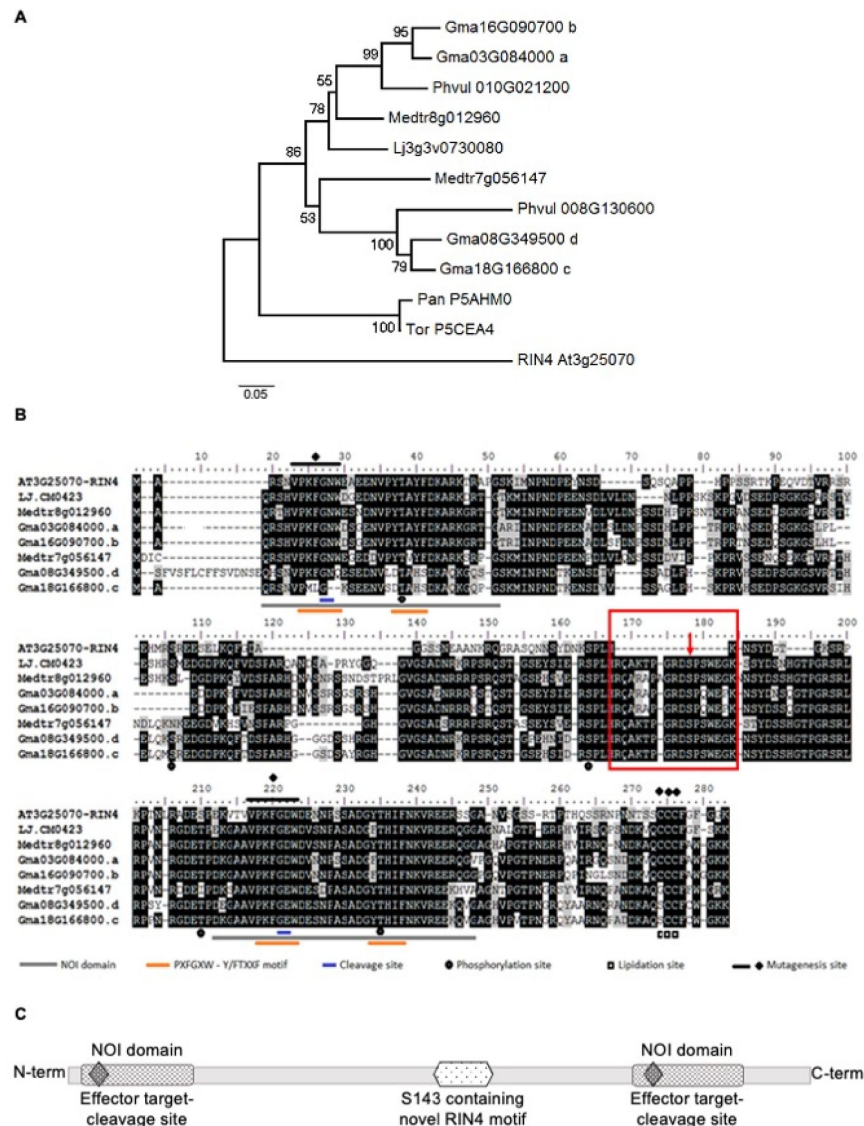


Fig. 1

Phylogeny of RIN4 protein homologs from model legume species and a novel-RIN4- motif within soybean RIN4

(A)

Phylogeny of soybean RIN4 closest homologs including GmRIN4a, GmRIN4b, GmRIN4c, GmRIN4d, *L. japonicus* RIN4 (Lj3g3v0730080), two *M. truncatula* RIN4 (Mtr8g012960, Mtr7g056147), two *P. vulgaris* RIN4 (Pvul010G021200, Pvul008G130600), and nodulating non-legume *P. andersonii* (Pan P5HM0) and its non-nodulating relative *Trema orientale* (Tor P5CEA4). Tree was rooted using *A. thaliana* RIN4 (At3g25070). (B) Novel-RIN4-motif within RIN4 protein sequences (red box). One of the identified phosphorylation sites, S143 is located within this motif (red arrow) within a "GRDSP" core sequence that is highly conserved among legume species and species of Rosales. Here, we show the sequence alignment of *Arabidopsis* RIN4 with soybean (Gma), *L. japonicus* (Lj), *M. truncatula* (Mtr), *P. vulgaris* (Pvul) and nodulating non-legume *P. andersonii* (Pan) and its non-nodulating relative *T. orientale* (Tor) RIN4 proteins. Grey underline indicates Nitrate-induced domain (NOI). Red underline indicates motif for proteolytic cleavage targeted by pathogenic effector proteins, while blue line shows cleavage site. The characteristic feature of the NOI-domain is that it harbors the PXFGXW motif which is the target site for effector protein. Tryptophan (W) is a crucial residue within the motif. RIN4c and RIN4d are missing W (blue arrow), and SMART analysis also predicted only one Pfam: AvrPt2 cleavage site. (C) RIN4 protein schematic based on soybean RIN4a/b. NOI domains at the proteins N- and C-terminus, diamond within NOI domain represents cleavage site targeted by effector protein(s), hexagon represents the Novel-RIN4-motif containing S143 phosphorylation site.

effector targeted cleavage sites are located within these motifs (36, [Fig. 1B](#), red underline; [Fig. 1C](#)). Our *in-silico* analysis found that all four soybean RIN4 proteins contain 2 NOI domains, whereas the AvrRpt2 cleavage site is absent in RIN4c and RIN4d N-terminal NOI domains ([Fig. 1B](#), blue arrow). This observation is based on sequence alignment ([Fig. 1B](#)) and prediction using the Simple Modular Architecture Research Tool (SMART).

Root hairs are the primary entry point for rhizobial infection in most legumes ([2](#)). We wanted to narrow the number of study subjects and, therefore, looked at the gene expression level of *RIN4* genes in soybean root hairs and roots using quantitative reverse transcriptase polymerase chain reaction (qRT-PCR) analysis. *RIN4a* and *RIN4b* showed much higher expression in root hairs than *RIN4c*, *RIN4d* and a *RIN4-like* gene (SI Appendix, [Fig. S3A](#)). *RIN4* transcripts levels were not induced upon rhizobial inoculation (SI Appendix, [Fig. S3A](#)), suggesting that the protein is regulated post-translationally, as was previously observed in other studies. In stripped roots (roots with root hairs removed), all *RIN4* genes displayed lower expression in comparison to root hairs, and none were up-regulated upon rhizobial treatment (SI Appendix, [Fig. S3A](#)). These results are consistent with *Arabidopsis* RNA-seq data revealing that *AtRIN4* is one of the most abundant transcripts in root hairs ([37](#)). Given that plant roots are constantly surrounded with microbes in the rhizosphere, it is not surprising that such a key immune regulator as RIN4 is being expressed and maintained at a high level. Our gene expression data were confirmed by Western-blot analysis using total protein extracted from root hairs and stripped roots (SI Appendix, [Fig. S3B](#)). RIN4 was detected using custom-specific anti RIN4 antibody generated against a mixture of RIN4a and RIN4b recombinantly expressed proteins. Anti-RIN4 antibody was tested using His-epitope tagged RIN4a, RIN4b, RIN4c and RIN4d recombinantly expressed and purified proteins. The RIN4 antibody recognizes only RIN4a and RIN4b, and not RIN4c and RIN4d (SI Appendix, [Fig. S3C](#)).

RIN4a and RIN4b share 93% identity on an amino acid level, whilst RIN4a/b share 64% identity with RIN4c and 62% identity with RIN4d on an amino acid level. RIN4c and RIN4d displayed lower expression in root hairs and roots compared to RIN4a and RIN4b (SI Appendix, [Fig. S3](#)). Furthermore, it was previously shown that only *RIN4b* complements *AtRIN4* in an *Arabidopsis rin4* mutant background ([33](#)). Therefore, we focused our attention on RIN4a and RIN4b in this study.

***RIN4a* and *RIN4b* are required for efficient nodulation**

Relatively few soybean mutants are available and, therefore, we searched for *rin4* mutants in the model legume *L. japonicus*, which forms determinate nodules, as does soybean. In *L. japonicus*, we could identify only one *RIN4* gene ([Fig. 1](#)). From the *L. japonicus* LORE1 mutant population database (<https://lotus.au.dk/>; 38, 39), we ordered two lines with exogenic LORE1 insertions in the *RIN4* gene locus (Plant ID 30000711, 30019656). Genotyping of the two exogenic lines did not reveal homozygous individuals (SI Appendix, [Table S2](#)), suggesting that *RIN4* homozygous mutation might be lethal in *L. japonicus*, as is the case in *A. thaliana* ([21](#)).

In order to assess the role of *RIN4* genes during the legume-rhizobium symbiosis, we targeted *RIN4a* and *RIN4b* by RNAi-mediated gene silencing. *RIN4a* and *RIN4b*-RNAi constructs were introduced into soybean roots via Agrobacterium-mediated hairy root transformation. Silencing of *RIN4a* and *RIN4b* resulted in significantly reduced nodule numbers (about 50-70% less) on soybean transgenic roots in comparison to transgenic roots carrying the empty vector control ([Fig. 2A and B](#)). Based on the qRT-PCR data ([Fig. 2C](#)), the transcript levels of both genes were significantly reduced, but not abolished, suggesting that both genes contribute to the formation of the symbiosis. Transcripts of the two genes are 92% identical, explaining the reduction in both transcripts ([Fig. 2C](#)). Because of the high-level identity of the two genes, it was very challenging to silence each gene separately.

To further confirm that soybean *RIN4a* and *RIN4b* play a role in formation of the nitrogen-fixing symbiosis, we targeted both genes using CRISPR-Cas9 gene editing technology to generate stable knock-out mutants. A CRISPR-Cas9 edited soybean knock-out line was obtained only in *RIN4b* in

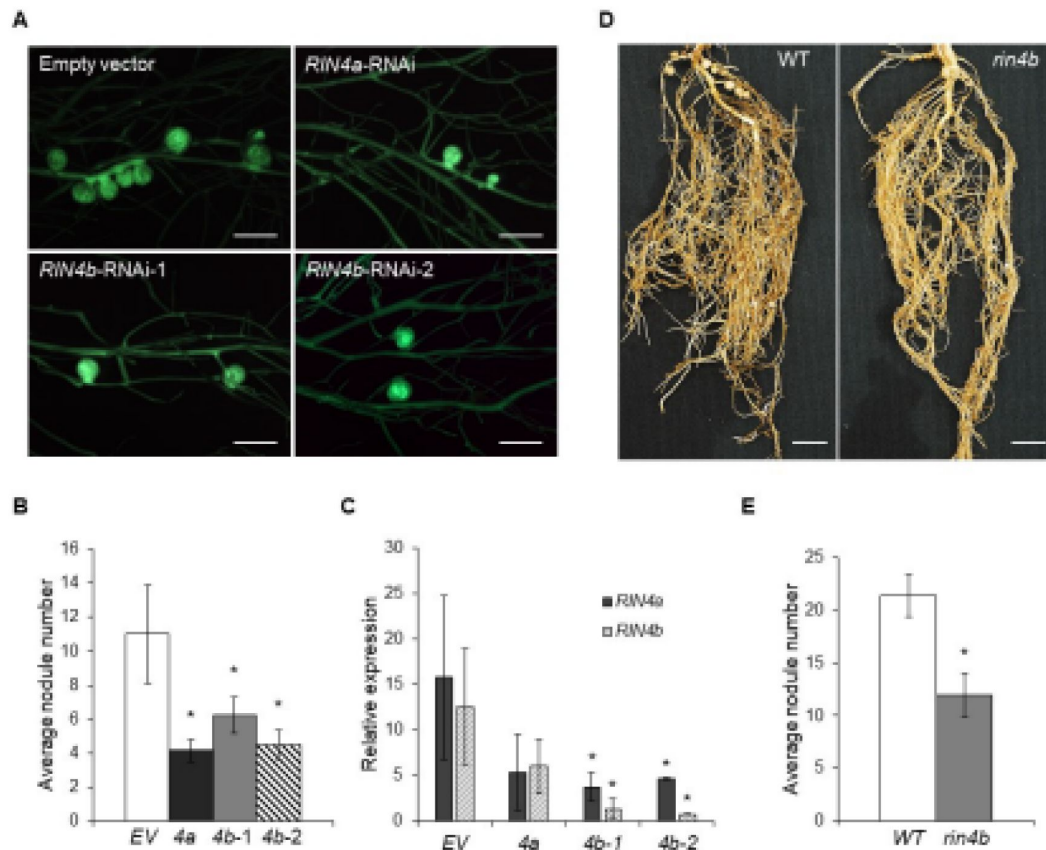


Fig. 2

***RIN4a* and *RIN4b* are required for proper symbiosis formation**

(A) Micrographs showing representative transgenic roots of gene silencing, visualized by green fluorescence originating from GFP marker carried by all vectors. Scale bars represent 2 mm. (B) Significantly reduced nodule numbers were observed on soybean transgenic roots carrying RNAi constructs targeting *RIN4a* and *RIN4b*. Roots were phenotyped 5 wpi. Representative data of one biological replicate, experiment was done in 3 biological replicates. Student's t-test * p<0.05. (C) Quantitative reverse transcription polymerase chain reaction (qRT-PCR) analysis confirmed reduced transcript levels of *RIN4a* and *RIN4b* in RNAi transgenic roots, strongly suggesting that the individual constructs targeted both genes. (D) Roots of wild-type *RIN4b* (*WT*) and mutant *rin4b* (in Bert background) homozygous mutant plants. Scale bars represent 2 cm. (E) Reduced nodule numbers in Bert soybean roots carrying a CRISPR-Cas9 edited 2 bp deletion in *RIN4b* (*rin4b*) in comparison to plants expressing wild-type *RIN4b* (*WT*) which out segregated as non-transgenic and non-edited. Experiments were done twice. Student's t-test * p<0.05.

the Bert cultivar background (40). This line contains a two base pair deletion within the second exon of the gene that results in a premature stop codon leading to reduced RIN4 protein abundance and significantly reduced *RIN4b* mRNA levels (SI Appendix, Fig. S4). This line produced significantly reduced nodule numbers in comparison to plants expressing wild-type *RIN4b* (Fig. 2D and E), supporting the findings of RNAi-mediated knock down and further confirming the role of *RIN4b* in the symbiosis. Hand-made cross-sections revealed that nodules on both wild-type and *rin4b* mutant roots were pink inside. The pink color reflects the presence of leghemoglobin, suggesting that the nodules were functional and fixing nitrogen. *rin4b*-CRISPR-Cas9 produced about 50-60% less nodules suggesting that there may be functional redundancy between *RIN4a* and *RIN4b*.

In our previous phosphoproteomic study, we found that RIN4 protein(s) phosphorylation occurred one-hour post-inoculation (hpi) (16). One of the identified phosphorylation sites is S143 (SI Appendix Fig. S5A). Intriguingly, S143 is harbored within the highly conserved “GRDSP” core motif located in the NRM (Fig. 1B, red arrow). The NRM is 100% identical between RIN4a and RIN4b (Fig. 1B). Given that the phosphorylation site is located within the NRM, we decided to further investigate the function of the S143 phosphorylation site. A pS143-specific peptide antibody was generated to detect phosphorylation (SI Appendix, Fig. S5A). We treated three-day old soybean seedlings with mock (H₂O) and wild-type *B. japonicum*, to confirm the phosphorylation previously observed. Root hairs (RH) were harvested 1 hpi separately from root tissue. AtRIN4 displays a basal phosphorylation level in mock-treated plants (26, 29); therefore, it is not surprising that our pS143 specific antibody shows RIN4 phosphorylation in mock-treated RH (SI Appendix Fig. S5B). ImageJ quantification of phosphorylation showed a 2-fold upregulation in response to *B. japonicum* when compared to mock-treated RH (SI Appendix Fig. S5C). To address the function of the identified S143 phosphorylation site in relation to the symbiosis, we introduced point mutation(s) by site-directed mutagenesis and generated RIN4a^{S143A} and RIN4b^{S143A} mutated proteins that cannot be phosphorylated at S143. Aspartic acid (D) was introduced to mimic the phosphorylation status and phospho-mimic mutant versions (RIN4a^{S143D} and RIN4b^{S143D}) were created. RIN4a and RIN4b were N-terminally tagged with HA-epitope to detect the presence of the introduced mutant and native versions of the protein in soybean transgenic roots. Interestingly, ectopic expression of RIN4a^{S143A} and RIN4b^{S143A} mutant proteins led to significantly reduced nodule numbers on transgenic roots in comparison to roots expressing wild-type RIN4b and empty vector control, whereas expression of RIN4a^{S143D} and RIN4b^{S143D} did not affect nodule numbers (SI Appendix, Fig. S6). These results suggest that phosphorylation at the S143 site in RIN4a and RIN4b is required for efficient symbiosis formation. In order to verify protein expression in transgenic roots, Western-blot analysis was performed on total protein extracts from transgenic roots. All constructs displayed two protein bands (phosphorylated and non-phosphorylated versions) at the expected size (around 30 kDa) showing that the introduced versions of RIN4a and RIN4b were expressed (SI Appendix, Fig. S6), contributing to the observed phenotype.

Phosphor-negative RIN4b^{S143A} does not complement *rin4b* mutant

Complementation of the *rin4b*-CRISPR-Cas9 mutant was carried out by introducing either RIN4b, RIN4b^{S143A}, RIN4b^{S143D} or empty vector (control) via *Agrobacterium*-mediated hairy root transformation into soybean transgenic roots induced on *rin4b* mutant plants. This experiment confirmed that RIN4b^{S143} is critical for RIN4b symbiotic function, as the phosphor-negative mutant version (RIN4b^{S143A}) of the protein, similar to the empty vector, was unable to rescue the nodulation phenotype observed on the mutant plants (Fig. 3A and B). In contrast, transgenic roots expressing RIN4b or the phosphomimic version RIN4b^{S143D} restored nodulation in comparison to transgenic roots carrying empty vector (Fig. 3). Therefore, we refer to S143 as nodulation-related S143 in the following parts of the manuscript. Expression of the HA-tagged RIN4b wild-type and mutant proteins in the transgenic roots was confirmed via Western-blot analysis (Fig. 3C).

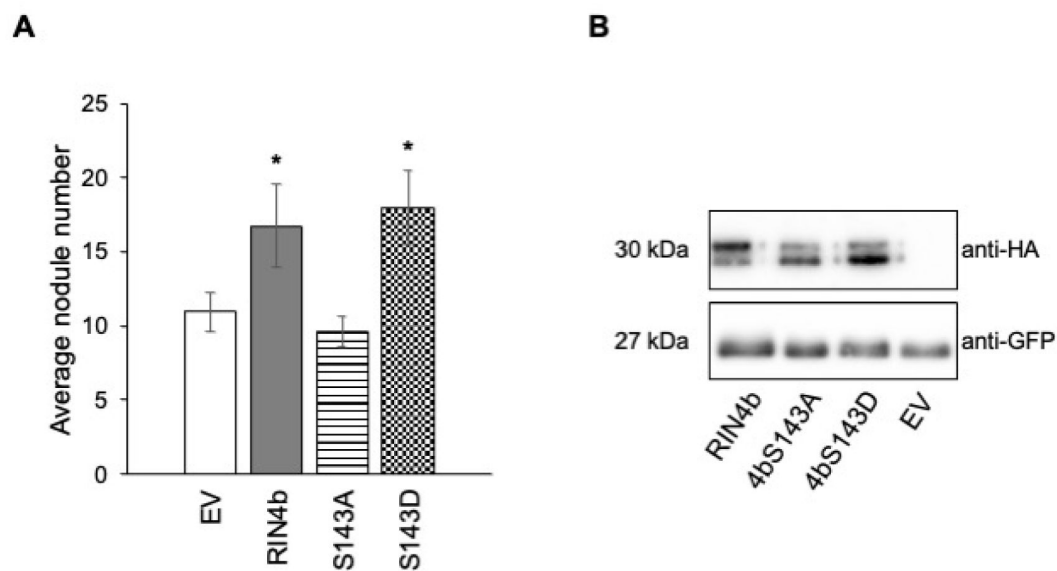


Fig. 3

Phosphor-negative RIN4b^{S143A} does not complement *rin4b* mutant phenotype

(A) *rin4b*-CRISPR-Cas9 mutant was transformed with RIN4b, RIN4b^{S143A}, RIN4b^{S143D} and empty vector. Only RIN4b and RIN4b^{S143D} could rescue the phenotype caused by *rin4b* mutation (micrographs on the right). RIN4b^{S143A} and an empty vector could not restore nodule numbers on the transgenic roots (micrographs on the left). Micrographs showing representative transgenic roots expressing the respective constructs, visualized by green fluorescence originating from GFP marker. Scale bar represents 1 mm. (B) Graphical representation of the results from 2 biological replicates. Error bars represent standard error. Student t-test, * $p < 0.05$. (C) Western-blot analysis showed that the transgenic roots expressed the HA-tagged RIN4b proteins, as well as the GFP marker carried by the vector.

RIN4a and RIN4b closely associate with symbiotic receptor-like kinases NFR1 α and SymRK β *in planta*

There are two active kinases required for early signal transduction during legume symbiosis development, NFR1 and SYMRK ([11](#), [9](#)). In soybean, both NFR1 and SymRK are present in two copies (NFR1 α/β and SymRK α/β ; [6](#), [8](#)). Ectopic over-expression of NFR1 α led to increased nodule numbers on transgenic roots, whereas this phenotype was not observed when NFR1 β was overexpressed ([6](#)). RNAi-mediated gene silencing was performed on both soybean *SymRK* genes. Silencing of *SymRK β* showed a stronger phenotype suggesting that this protein has the major function in nodulation ([8](#)). Since NFR1 α and SymRK β seem to be the major players in soybean, we decided to investigate the interaction of RIN4a and RIN4b only with NFR1 α and SymRK β and not their homologs (NFR1 β and SymRK α). Previously it was shown that SymRK in *L. japonicus* undergoes proteolytic cleavage, when the Malectin-like domain (MLD) within the protein's extracellular region is cleaved off creating SymRK Δ MLD ([10](#)). Δ MLD was easier to detect when expressing both full-length (FL) SymRK β and SymRK $\beta\Delta$ MLD in tobacco leaves. Therefore, for future experiments we used the SymRK $\beta\Delta$ MLD construct, instead of SymRK β -FL. In the *in planta* Bimolecular Fluorescence Complementation assay (BiFC), RIN4a and RIN4b showed interaction with each other as previously shown ([33](#)) and were used as a positive control for the assay (SI Appendix, **Fig. S8A**). Co-expression of RIN4a/b with SymRK $\beta\Delta$ MLD as well as NFR1 α resulted in YFP fluorescence detected by Confocal Laser Scanning Microscopy (SI Appendix, **Fig. S8**) suggesting that RIN4a and b proteins are closely associated with both RLKs. No interaction could be observed between RIN4a/b and the P2K1 receptor, which was used as negative control in the BiFC assay (SI Appendix, **Fig. S8** D and G). The reconstituted YFP signal localizes to the plasma membrane (PM) in accordance with earlier observations where the investigated proteins were localized to the PM ([10](#), [11](#), [33](#)). PM was visualized by application of a membrane staining dye, FM4-64 (SI Appendix, **Fig. S8**). The interaction between RIN4a/b and NFR1 α and SymRK $\beta\Delta$ MLD was confirmed in a Split-Luciferase Complementation assay (SI appendix, **Fig. S9**), when the proteins were fused to the N-terminal and C-terminal domains of the Luciferase enzyme, which upon reconstitution results in bioluminescence.

SymRK β phosphorylates the nodulation-related S143 phospho-site *in vitro* and *in planta*

The kinase activity of soybean NFR1 α was demonstrated by Choudhury and Pandey ([41](#)). Here, we show that soybean SymRK β possesses an active kinase domain (SymRK β KD, SI Appendix, **Fig. S10**). The isolated soybean SymRK β -KD showed strong autophosphorylation activity, as well as the ability to trans-phosphorylate MBP, a universal substrate (SI Appendix, **Fig. S10**). Furthermore, to confirm that phosphorylation was triggered by SymRK β -KD, we introduced a point mutation in the same position as previously described by Yoshida and Parniske ([41](#)) for the *L. japonicus* SymRK kinase domain. Specifically, D734N (which corresponds to D738 in LjSymRK) was introduced, in the activation loop, and led to inactivation of the SymRK β kinase activity (SI Appendix, **Fig. S10**).

Both NFR1 α and SymRK β kinase domains phosphorylated RIN4a and RIN4b *in vitro* when radioactive [γ -32P] ATP was used to visualize the phosphorylation (SI Appendix, **Fig. S11** A).

In order to ascertain which of these two kinases phosphorylates the nodulation-related RIN4S143, we performed mass spectrometry-based Absolute Quantification (AQUA), a method that uses stable-isotope labeled peptides as internal standards to quantify proteins or post-translational modifications ([43](#)). The abundance of the heavy-labeled peptides and their corresponding endogenous peptides (peptides derived from native RIN4a and b) can be quantified using selected reaction monitoring mass spectrometry (MS-SRM). In our phosphoproteomic study of soybean root

hairs, we also observed phosphorylation of RIN4T173 (17 [↗](#)) which corresponds to the earlier published AtRIN4S141 triggered by bacterial flagellum epitope, flg22 (29 [↗](#)). Heavy-labeled AQUA peptides were generated against native peptides carrying S143 and T173 (to serve as a control), as well as phosphorylated versions of the peptides (SI Appendix, **Table S3** [↗](#)).

In vitro kinase assay was performed in the absence and presence of ATP (the donor of phosphate group), and MS-SRM was carried out to quantify phosphorylation levels of RIN4a^{S143}, RIN4b^{S143}, RIN4a^{T173} and RIN4b^{T173}. The nodulation-related S143 site was phosphorylated only by SymRKβ in the presence of ATP in RIN4a, as well as in RIN4b (SI Appendix, **Fig. S11** [↗](#) B), whereas phosphorylation of RIN4a^{T173} and RIN4b^{T173} was not detected (SI Appendix, **Fig. S11** [↗](#) B). No phosphorylation of RIN4a^{S143}, RIN4b^{S143}, RIN4a^{T173} and RIN4b^{T173} was observed when the proteins were co-incubated with NFR1α-KD (SI Appendix, **Fig. S11** [↗](#) B). Calibration curves were established and correlation coefficients were determined for both S143 and T173 peptides (SI Appendix Fig. S12). Peptides in both proteins were detected at a similar level when equal amounts were injected into the mass spectrometer (SI Appendix Fig. S12).

To further assess RIN4a and RIN4b phosphorylation by SymRKβ in an *in planta* environment, the proteins were co-expressed in Arabidopsis leaf protoplasts. Given that AtRIN4 lacks not only the S143 phosphorylation site but also the motif that carries this phosphorylation site, it provided clear evidence about the phosphorylation status of the protein induced by SymRKβ. RIN4a and RIN4b phosphorylation by wild-type SymRKβΔMLD is demonstrated on **Figure 4** [↗](#). SymRKβΔMLD kinase inactive version does not phosphorylate either RIN4a or RIN4b. Furthermore, it is also shown that phosphor-minus versions RIN4a^{S143A} and RIN4b^{S143A} do not display a phosphorylated band (**Fig. 4** [↗](#)), supporting the specificity of the α-pS143 phosphor-specific peptide antibody.

RIN4b acts at the intersection of symbiotic and immune signaling

In order to decipher the RIN4b contribution to symbiotic signaling, we performed gene expression analysis of well-known symbiotic signaling genes, such as the transcription factors (TF) NIN, NF-YA and ERN1 in the *rin4b* (CRISPR-Cas9) mutant background. *NIN* is a nodulation-specific TF, which was the first gene identified in the symbiotic pathway more than 20 years ago (44 [↗](#)). Both root epidermal and cortical signaling leads to activation of *NIN*. *NIN* contributes to bacterial infection in root hairs, to epidermal responses and cortical cell division leading to nodule organogenesis (15 [↗](#)). *NIN* activates Nuclear Factor-Y (NF-Y) transcriptional subunits, a heterotrimeric DNA-binding protein complex composed of NF-YA, NF-YB and NF-YC (15 [↗](#)). Soyano and colleagues (15 [↗](#)) identified *NF-YA* and *NF-YB* in *L. japonicus* as a target of *NIN* regulation. *LjNF-YA* overexpression caused changes in the root architecture, while overexpression of *LjNF-YB* did not show root alterations, therefore *LjNF-YA* was designated as the primary player in cortical cell division (15 [↗](#)). *ERN1* in *L. japonicus* and *M. truncatula* encodes an AP2/ERF transcription factor. *ERN1* is the central regulator of the infection process and is directly regulated by the CCaMK-CYCLOPS complex (45 [↗](#)).

Hayashi and colleagues (46 [↗](#)) identified four *NIN*-like genes in soybean (*GmNIN1a*, *GmNIN1b*, *GmNIN2a* and *GmNIN2b*). Since *NIN1b* was detected at a low level and did not display significant induction in response to rhizobia (46 [↗](#)), only *GmNIN1a*, *GmNIN2a* and *GmNIN2b* were included in our analysis. While it was shown that both *GmERN1a* and *GmERN1b* responded to rhizobial inoculation, in our hands only *ERN1a* could be amplified. Sequence alignment identified at least three NF-YA transcription factor homologs (NF-YA 1, 2, 3) in soybean. Based on preliminary experiments, only *NF-YA1* and *NF-YA3* were found to be expressed and, therefore, *NF-YA2* was excluded from the analysis. For these experiments, *rin4b*-CRISPR-Cas 9 mutant in the Bert background (M4 and M5 bulk) and Bert wild-type seedlings were used. Seedlings four days post-germination were treated with mock (H₂O) and *B. japonicum*. Given the limited number of *rin4b* mutant seeds, root hair separation from roots was not possible, therefore entire roots were

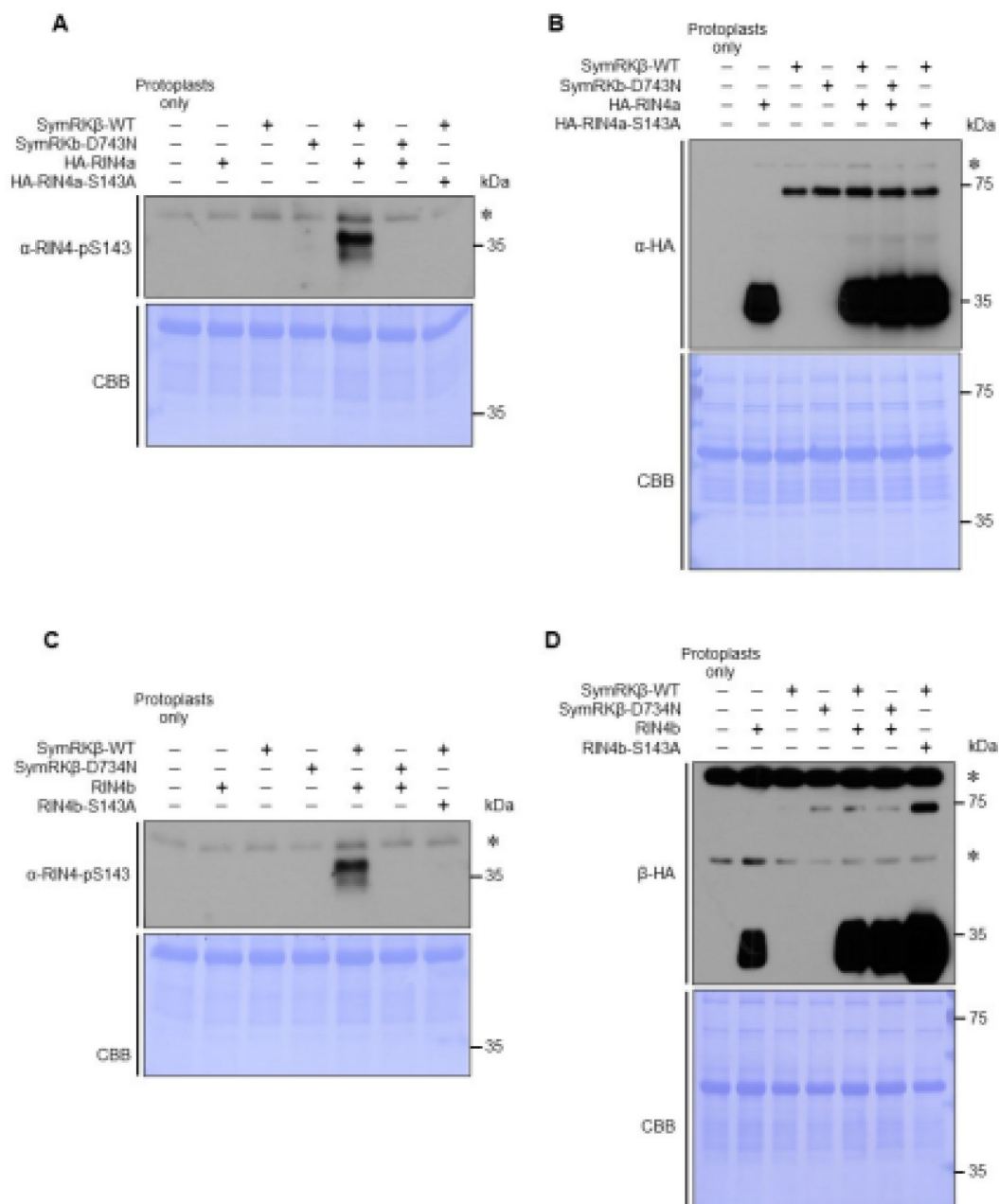


Fig. 4

Nodulation-related S143 is phosphorylated by SymRKβ *in planta*

(A) Phosphorylation of RIN4a by SymrkβΔMLD is demonstrated using α-pRIN4-S143, while no phosphorylation can be observed when RIN4a was co-expressed with the kinase inactive version of SymRKβ *in planta*, in *Arabidopsis* protoplasts. (B) shows the expression of all components co-expressed in protoplasts: HA-tagged RIN4a, HA-RIN4aS143A, HA-tagged wild-type SymrkβΔMLD and kinase-dead SymrkβΔMLD. (C) Phosphorylation of RIN4b by SymrkβΔMLD is demonstrated using α-pRIN4-S143, while no phosphorylation can be observed when RIN4b was co-expressed with the kinase inactive version of SymRKβ (D) shows the expression of all components co-expressed in protoplasts: HA-RIN4b, HA RIN4bS143A, wild-type HA-SymrkβΔMLD and HA-SymrkβΔMLD-D734N. *: non-specific band shows equal loading of the protein extracts as well as CBB staining of the membranes displays equal protein extract loading.

harvested one hour and 12 hours post-inoculation. *NIN1a* expression (the closest homolog to *LjNIN* and *MtNIN*; 46), in agreement with previously published reports, was induced in wild-type plants 12 hours post-inoculation with

B. japonicum (Fig. 5), whereas its expression in *rin4b* mutant was not induced. *NIN2a* and *NIN2b* showed induction in *rin4b* upon rhizobial treatment, but at a significantly lower level in comparison to wild-type roots (Fig. 5). Since NIN is responsible for induction of NF-YAs, their reduced expression in *rin4b* plants was not unexpected (Fig. 5). *ERN1* expression is also affected in *rin4b* mutant plants, this data suggest that RIN4b is located upstream of the CCaMK-CYCLOPS complex, as *ERN1* expression is CYCLOPS dependent (47). Based on the results of the expression data presented above, it seems that both *NIN*-dependent and *NIN*-independent branches of the symbiotic signalling pathways are impacted in the *rin4b* mutant background. This indicates that the role of RIN4 protein in the symbiotic pathway can be placed upstream of CYCLOPS, as the CYCLOPS transcription activating complex is responsible (directly or indirectly) for the activation of all TFs tested in our expression analysis (48, 47).

Since the expression of all investigated transcription factors was impaired in the *rin4b* mutant background in comparison to Bert wild-type plants at 12 hpi (Fig. 5), the data suggest that the absence of *RIN4b* negatively affects the symbiotic signaling pathway. While the expression of these marker genes was reduced in the *rin4b* mutant plants, some induction upon inoculation was still apparent. We attribute these findings to the functional redundancy between *RIN4a* and *RIN4b*.

As an invading organism, rhizobia also elicit plant immune responses (48, 49), although transient in the case of compatible interactions. These responses need to be amplified or suppressed depending on whether the host senses the bacteria as friend or foe. RIN4 is an immune regulator that is a key intersection between PTI, effector-mediated defense suppression and NTI (24). The work presented here also supports a role for RIN4 as a key determinant in symbiotic signaling.

In Arabidopsis, absence of RIN4 enhances PTI responses. Overexpression of AtRIN4 leads to PTI inhibition as no callose deposition was observed when plants were treated with flg22 (22). The function of AtRIN4 is controlled by the specific phosphorylation status of the protein. In the case of soybean nodulation, silencing of *rin4* expression, mutagenesis (i.e., *rin4b*-CRISPR-Cas9 mutant) or disruption of S143 phosphorylation resulted in a significant reduction in nodulation. Absence of AtRIN4 leads to increased PTI responses and, therefore, it might be that GmRIN4b absence also causes enhanced PTI which might have contributed to significantly less nodules. However, this may be too simplistic an idea given the impact of *rin4* mutagenesis on nodule-related gene expression, as well as the lack of knowledge of other likely partners (e.g., interacting R proteins) that might also be playing a role. Compared to leaves there is also a paucity of data as to how plant roots (51), especially with regard to soybean, respond to pathogen infection and elicitation, as well as any functional role for RIN4.

While no redundancy was shown between RIN4a and RIN4b in plant-pathogenic interactions, our data suggest functional redundancy between the two isoforms in the symbiosis, pointing toward a likely scenario when immune responses in the aerial part differ from immune responses in the root. Several defense-responsive genes which are well described in leaf immune responses could not be confirmed to respond in a similar manner in root (51). It was also shown that plant immune responses in roots are compartmentalized and specialized (51). Clearly, there remains much to be discovered as to how RIN4, as well as other components of both the pathogen response and symbiotic response pathways, converge to ultimately distinguish friend from foe in the root.

In soybean, an effector-mediated hijacking of the symbiotic signaling revealed nodule formation in the absence of the NFR1 receptor. NF-deficient *Bradyrhizobium elkanii* was able to elicit nodule formation on *nfr1* mutant plants, whereas NF-deficient T3SS (Type III secretion system is required

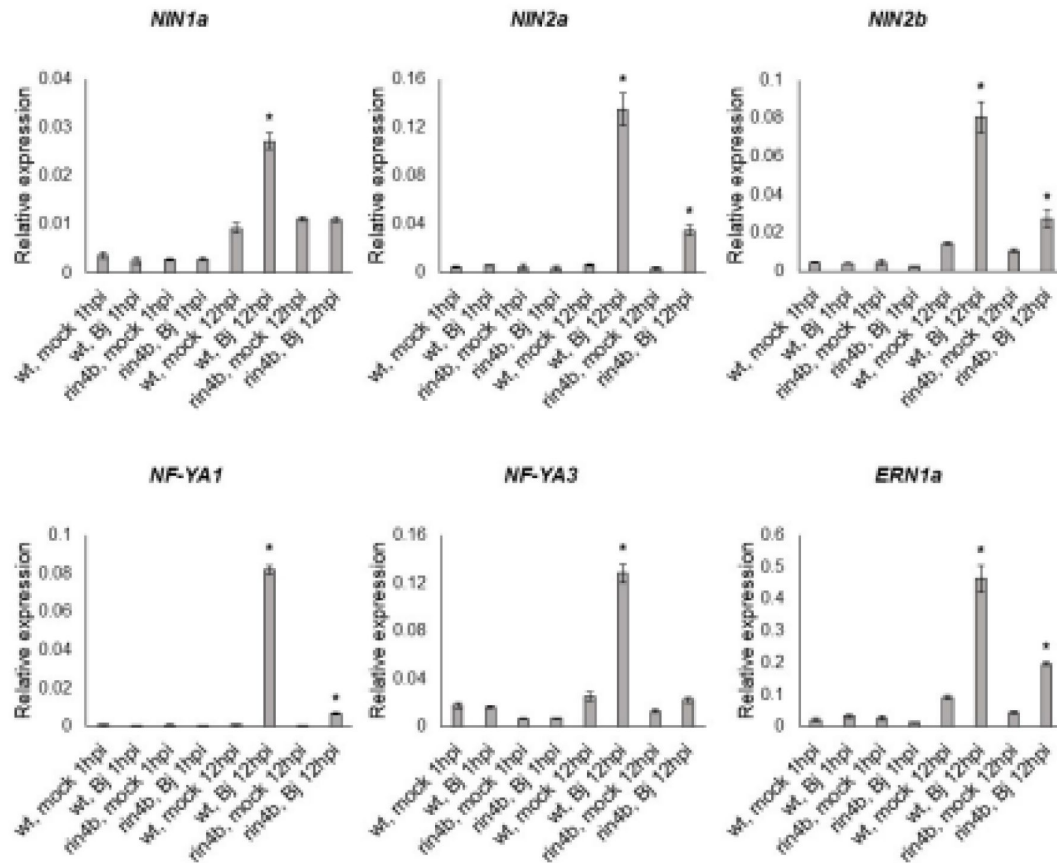


Fig. 5

Symbiotic signaling is affected in the *rin4b*-CRISPR-Cas9 mutant

Expression of transcription factors (TF) involved in the symbiotic signaling pathway was investigated in the *rin4b* mutant background by qRT-PCR. Induction of the closest homolog of nodulation-specific transcription factor *NIN1a* can be observed in wild type (wt) and it does not change in *rin4b* in response to *B. japonicum* (Bj) 12 hpi. While two other homologs (*NIN2a* and *NIN2b*) respond to Bj in *rin4b*, their induction is significantly lower in comparison to Bert wt. Two soybean *NF-YA* TFs were tested as they are activated by NIN, therefore their lower expression in response to Bj in *rin4b* was expected. *ERN1* is another TF downstream of NIN. Its induction in *rin4b* in response to Bj is significantly lower than the response in Bert wt. Student's t-test * $p < 0.05$.

for effector proteins delivery) double mutant was not able to induce nodule formation (52 [DOI](#)), suggesting that effector protein(s) are required for successful nodulation in soybean. Taking into consideration that rhizobial effector(s) were able to mediate nodule formation and AtRIN4 up-regulated phosphorylation is a target for an effector protein, our proposed model of RIN4 function in the symbiosis is the following: GmRIN4^{S143} is phosphorylated by SymRK, a receptor whose structure is similar to known pattern recognition receptors (PRR) (though no ligand was shown for SymRK so far). Putative effector proteins injected by compatible rhizobium recognize phosphorylated S143, ensuring downstream symbiotic signaling perhaps via effector-mediated interaction of RIN4 to modulate basal immune response. Another likely scenario would be phosphorylation mediated interaction of RIN4a and RIN4b proteins with other downstream proteins which would require further investigation into protein complex formation upon rhizobial infection.

Taken together, our results indicate that successful development of the root nodule symbiosis requires cross-talk between NF-triggered symbiotic signaling and plant immune signaling mediated by RIN4. Based on data presented here, all this likely happening upstream of the transcriptional regulator CYCLOPS, a further piece of evidence (in addition that GmRIN4a/b phosphorylation was detected only in root hairs, where the infection takes place) that involvement of soybean RIN4 in nodulation is required early on in the signalling pathway. Furthermore, it is likely that soybean RIN4 phosphorylation by SYMRK at S143 modulates immune responses to enable symbiotic interaction with rhizobia.

Our data present important findings on the way to unraveling how legume plants prioritize beneficial interactions and suppressing immunity-related signals to enable symbiotic associations.

Material and Methods

A detailed description of the methods used in this study can be found in SI Appendix, Supplementary Materials and Methods.

Seedling growth, treatment and total root, root hair and stripped root collection

In order to obtain root hairs and stripped roots, and total roots for protein and RNA extraction, *Glycine max* cv. Williams 82, and *rin4b*-CRISPR-Cas9 in Bert background M4 and M5 seeds, and Bert wild-type seeds were used. Seeds were surface sterilized with 20% household bleach, left in 20% bleach for 10 min, then washed three times with sterile diH₂O. Afterwards, the seeds were incubated for 10 min in 0.05% HCl and subsequently washed four times with sterile diH₂O before sowing onto 1% water-agar plates (20 cm in diameter glass Petri dish, MG Scientific, WI, USA). Williams 82 seeds (used for root hair experiments) were germinated in a growth chamber (Convion Growth Chamber PGR15) with 90% humidity at 28°C in the dark and three days old seedlings were used for treatment. *rin4b* and Bert wild-type seeds were germinated at room temperature (21-23°C) in the dark. Seedlings were spray-inoculated with an OD₆₀₀ ~ 0.2 of *B. japonicum* USDA110 wild type, and sterile water sprayed as mock. Root hairs were harvested 1 hpi by flash-freezing in liquid nitrogen, followed by stirring for 15 min, which shears the root hairs from the roots. Root hairs were then isolated by filtration and both root hairs and stripped roots (i.e., roots with root hairs removed) stored separately at -80°C until further processing. As for *rin4b* and Bert wild-type plants, roots were inoculated 4 days post sowing with mock and *B. japonicum* USDA 110 with an OD₆₀₀ ~ 0.2, and entire roots harvested 1 hpi and 12 hpi, and stored at -80°C until further processing.

Protein purification from soybean root hairs, roots and transgenic roots

Soybean roots were ground in liquid nitrogen using mortar and pestle. Ground tissue was transferred to an Eppendorf tube and 750 μ l of extraction buffer (0.9 M Sucrose, 100 mM Tris HCl, pH 8.0; 10 mM EDTA, 0.4% β -mercaptoethanol, 0.2% Triton-X 100, Plant protease inhibitor, P9599 and phosphatase inhibitor, P0044 from Sigma) was added. Samples were vortexed and incubated on ice for 10 min. Equal volume of Tris buffered phenol was added, samples were vortexed and incubated for at least 1 h at 4°C rotating. Samples were centrifuged for 6 min at 13000 rcf at 4°C. Upper phase was transferred to a new tube and one ml ice-cold precipitation buffer (0.1 M Ammonium acetate dissolved in high quality methanol) was added and proteins were precipitated overnight at -20°C. Next day, samples were centrifuged for 10 min at 3500 rcf at 4°C. Pellet was resuspended in 1 ml ice-cold precipitation buffer, vortexed and centrifuged (10 min, 3500 rcf, 4°C). This step was repeated. Afterwards, the pellet was washed with 80% ice-cold acetone, vortexed and centrifuged (10 min, 3500 rcf, 4°C), this step was repeated twice. The pellet was resuspended in 70% EtOH, vortexed, centrifuged and air dried. Proteins were resuspended in 8 M Urea (solubilized in 50mM Tris-HCl, pH 8.0) and protein concentration was measured using Pierce 660 nm Protein Assay (Thermo Scientific, Rockford, IL, USA) and used for SDS-PAGE and subsequent Immuno-blotting analysis.

SDS-PAGE and Immuno-blot analysis

10% or 12% Tris-PAGE gel was prepared without SDS (0.375 M Tris, pH 8.8; 10-12% Bis Acrylamide (40%, 29:1), 0.1% APS, TEMED) to separate proteins and for subsequent Western blot assay. Proteins were transferred onto a PVDF membrane at 4°C over-night at 30 V. After transfer, the membrane was incubated in 1% BSA (Gold Biotechnology, St. Louis, MO, USA) in TBS-T (with 0.1% Tween-20 from Fisher Scientific) or 5% milk (fat-free skim milk, SACO Foods, Middleton, WI, USA) in TBS-T. Custom RIN4 antibody was generated by AnaSpec (Fremont, CA, USA) and used at 1:5000 dilution in 5% Milk. Custom RIN4 phosphorylation site specific peptide antibody, α -pS143 (GRDPSPQWEPKNSYD) were generated by GenScript (Piscataway, NJ, USA) and used at 1:4000 dilution in 1% BSA to detect proteins isolated from soybean root hair. Secondary rabbit HRP conjugated antibody was used in 1:10000 dilution and obtained from Jackson ImmunoResearch (West Grove, PA, USA).

HA-tagged RIN4a and RIN4b expressed in soybean transgenic roots were detected using monoclonal HA-antibody (Roche Diagnostics, Germany) with or without HRP-conjugate and used in 1:2500 dilution. 1:5000 dilution was applied to detect HA-tagged RIN4a/b expressed in protoplasts. In case of non-conjugated HA antibody, secondary rat antibody was used in 1:10000 dilution in 5% milk. Signal was visualized using 1:8 Femto:Pico Pierce SuperSignal chemiluminescence substrate (Thermo Fisher Scientific, USA). GFP expressed in transgenic roots was detected using anti GFP antibody (Invitrogen, USA) at 1:5000 dilution in 5% milk. To detect RIN4(a/b) S143 phosphorylation in protoplasts, membrane(s) were blocked in a 2% BSA (dissolved in TBS-T) solution, and were incubated with α -pS143 at 1:3000 in 2% BSA solution with subsequent incubation with secondary rabbit HRP conjugated antibody at 1:15000 dilution. Pictures were taken using a UVP Camera (BioSpectrum 815 Imaging System; Upland, CA, USA) system, or x-ray film was exposed by the blot and photographically developed. Images were inverted and brightness and contrast were adjusted during figure preparation.

Targeted Multiple Reaction Monitoring Mass Spectrometry (MRM-MS)

Custom-made stable-isotope-labeled AQUA peptides carrying the S143 and T173 phosphorylation sites (RIN4a and RIN4b are identical, IS Appendix, [Table S3](#)) were generated by Sigma-Aldrich (The Woodlands, TX, USA). The peptides were resuspended in 50% Acetonitrile (ACN) to generate

10 pmol/ μ l stock solutions stored at -80°C . Working solutions of the peptides were diluted in 5% ACN with 0.1% formic acid (FA). After 30 min *in vitro* kinase assay (described above), the samples were spun down and a mixture of heavy-labeled S143, pS143 (phosphorylated version of S143 peptide), T173 and pT173 were added (100 fmol of S143 and 50 fmol of T173) to the samples in 20 μ l reaction volume prior to digestion. In solution trypsin digestion was performed overnight at 37°C as follows: 2.5 μ l of reduction solution (100 mM Ammonium bicarbonate/ABC with 100 mM DTT) was added, and samples were incubated at 37°C for 30 min; 2 μ l alkylation solution (0.5 M Iodoacetamide dissolved in 100 mM ABC) was added to the samples and samples were incubated up to 1 hour at room temperature in the dark. Samples were neutralized by 80 μ l 10 mM DTT in 10 mM ABC solution, before 10 μ l sequencing grade modified Trypsin (100 ng/ μ l) from Promega (Madison, WI, USA) dissolved in 100 mM ABC was added. After overnight trypsin digestion, samples were centrifuged for 2 min at 13 000 rcf, frozen in liquid nitrogen and lyophilized. Samples were dissolved in 0.1% FA. To quantify phosphorylation, we used Thermo-Scientific Quantiva triple Quadrupole mass spectrometer coupled to an Eksigent 1D plus (SCIEX) Nano-LC (liquid chromatography) instrument. 20 cm long column of 75 μ m in diameter filled with HxSIL-C18 (particle size 5 μ m, Hamilton Company, Reno, NV, USA) was used for sample separation over a 25 min gradient run. 10 μ l sample per injection was used in three technical replicates for each biological replicate. MS-MRM (in positive ion mode) was run over 25 min acquisition time at 3 mTorr CID gas pressure, defined collision energy (V) and cycle time (1 sec) for at least three transitions for each peptide (IS Appendix, [Table S2](#)). MS RAW data files were processed using Skyline software (MacCoss Laboratory Software, University of Washington, Seattle, WA, USA; <https://skyline.ms/project/home/begin.view>) to obtain the area under curves for integrated LC-SRM peaks. Integrated peaks were manually inspected to ensure all quantified transitions had overlapping retention times. Native peptide abundance is expressed as the ratio of endogenous peptide to labeled standard peptide expressed in percentage. MS-SRM was performed at the Gehrke Proteomics Center of the University of Missouri.

Materials and Methods

Sequence alignment and phylogenetic analysis of RIN4 protein homologs

Putative RIN4 homologs were defined with at least one NOI domain (PFAM database ID: PF05627 called AvrRpt-cleavage family) in protein sequences. Protein sequences were downloaded from the Phytozome database (<http://phytozome.jgi.doe.gov/>), (<http://www.kazusa.or.jp/lotus/>), NCBI database (1, <https://www.ncbi.nlm.nih.gov/>), PeanutBase (<https://www.peanutbase.org/home>), *Lotus japonicus* database (<https://www.kazusa.or.jp/lotus/>) and the UniProt protein database (<https://www.uniprot.org/>), from GigaDB (34, <https://www.re3data.org/repository/r3d100010478>) and from 1kp project (2, <https://db.cngb.org/onekp/>). All predicted RIN4 homologs were confirmed with at least one NOI domain using Simple Modular Architecture Research Tool (SMART, <http://smart.emblheidelberg.de/>) and manual check. A full-length alignment of all putative RIN4s were made using the MAFFT server tool (<http://mafft.cbrc.jp/alignment/server/>) with iterative refinement methods (E-INS-i), multiple conserved domain alignment option, scoring matrix BLOSUM62, default gap opening penalty 1.53.

The alignments were used to generate phylogenetic trees. The phylogenetic tree in the [Fig. 1A](#) was generated using the Neighbor Joining (NJ) method (3) with the JTT substitution model (4) and 1000 bootstrap resampling value. The phylogenetic tree in the [Fig. S1](#) was generated using the Average linkage (UPGMA) method and the JTT substitution model. Phylogenetic tree was visualized using the FigTree tool.

Composite plant generation and nodulation assay

In order to generate soybean composite plants expressing the respective constructs, *Agrobacterium*-mediated hairy root transformation was performed as described in Tóth *et al.*, 2016 (5) with the following modifications. Hairy root induction was initiated by poking the plants with a needle tip (BD PrecisionGlide Needle, 23G x 1½ TW IM/0.6mmx40mm, sterile, Franklin Lakes, NJ, USA) carrying the respective *agrobacterium*. Before planting the composite plants into sterile potting mix (3:1 vermiculite: perlite mix; Hummert International, Earth City, MO, USA), *Agrobacterium* induced roots were subjected to fluorescence microscopy. The vectors used in this study contain a constitutive GFP marker in order to identify transgenic roots. Non-transgenic roots were removed before planting. 24–48 hours after planting, plants were inoculated with *B. japonicum* USDA 110 wild type at an OD600 ~ 0.05. Composite plants were grown in a walk-in growth chamber (Conviron GR 64; non-controlled humidity; 16 h/light/26°C and 8 h/dark/24°C). The nodulation phenotype was observed at 5 wpi. Leica M205 FA stereomicroscope was used to take pictures of the nodulated transgenic roots at 8.0 x magnification.

In order to determine the nodulation phenotype of *rin4b* mutant plants, seeds (M4) of *rin4b* homozygous CRISPR-Cas9 lines and non-edited Bert plants, with native *RIN4b* (originating from plants that went through transformation but segregated as non-transgenic and non-edited), were surface sterilized and seedlings transferred to perlite-vermiculite potting mix and were grown in controlled walk-in growth chamber (Conviron GR 64; non-controlled humidity non-controlled humidity; 16 h/light/23°C and 8 h/dark/21°C). Plants were inoculated with *B. japonicum* USDA 110 wild type at an OD600 ~ 0.05 and subjected to phenotyping 5 wpi. A Panasonic (Lumix) camera was used to take pictures of Bert wild-type and *rin4b* mutant nodulated roots.

CRISPR-Cas9 edited *RIN4b* (Glyma16G090700) mutant generation

The transformation vector was constructed using a modified version of the *Glycine max* codon optimized Cas9 plasmid from Michno *et al.* (6, Addgene plasmid # 59184). The pBlu guide RNA shuttle vector used was identical to the one described in Michno *et al.* (6) (addgene #59188). *RIN4b* target was selected using Stupar lab's CRISPR design tool (<http://stuparcrispr.cfans.umn.edu/CRISPR/>). Guide RNA target oligos were designed based on the program then synthesized and annealed in a 10X PCR buffer at 50 °C for six hours. The pBlu/gRNA shuttle vector was digested using BbsI (New England Biolabs # R0539S) following manufacturer's guidelines. Digested product was run on an agarose gel and extracted using Qiagen gel extraction kit. The target oligos and shuttle vector were ligated using T4 ligase (New England Biolabs # M0202S). The ligation product was transformed into DH5 alpha (Life technologies) competent cells. The pBlu vector containing the target oligos and the destination vector were both digested with EcoRI (New England Biolabs #R0101S). The gRNA cassette and digested destination vector were ligated using T4 ligase, and transformed into DH5 alpha competent cells. *RIN4b* CRISPR-Cas9 construct was delivered into Bert-MN-01 background using K599 *Agrobacterium* (disarmed strain 18rv12). Methods for delivery and growth of whole-plant transformants were performed as previously described

(7). T0 plants and subsequent progeny were tested for the presence of the CRISPR-Cas9 transgene using PCR with transgene-specific primers followed by agarose gel electrophoresis. Mutations in transformed plants were screened as previously described using cleaved amplified polymorphic sequences (8) and/or PCR heteroduplex (9, 10) analysis. The mutated alleles in specific plants were confirmed by Sanger sequencing of PCR amplicons. Plants harboring deletions and no transgene were selected for further analyses.

Cloning

GmRIN4a (Glyma.03G084000.1) and *GmRIN4b* (Glyma.16G090700.1), *GmSymRKβ* (Glyma.09G202300.1), *GmNFR1α* (Glyma.02G270800.1) CDS were cloned via Gateway BP reaction into pDONR/Zeo entry vector. Sequence accuracy was confirmed by Sanger sequencing performed by MU DNA Core facility. These entry clones were used to fuse the haemagglutinin (HA) epitope onto the N-terminus of RIN4a and RIN4b and to introduce point-mutations by site-directed mutagenesis. HA-epitope tagged entry clones were used for subsequent cloning into a modified, gateway compatible pCambia vector (11 [↗](#)) for ectopic expression in soybean composite plants. 120-150 bp of *RIN4a* and *RIN4b* transcripts in the 3'UTR regions were cloned into the pDONR/Zeo entry vector. These clones were used for subsequent cloning into a modified gateway compatible pCambia vector for RNAi-mediated gene silencing (12 [↗](#)). Both modified pCambia vectors contain a GFP marker for transgenic root identification. pDONR/Zeo: RIN4a/b and pDONR/Zeo: SymRKβ/NFR1α entry clones were used to create constructs for BiFC, Split-Luciferase, as well as protoplast (protein expression and phosphorylation) experiments.

Bimolecular Fluorescence Complementation Assay

pDONR/Zeo:*RIN4a* and *RIN4b* were used to create N-terminally fused split YFP constructs in the pAM-PAT vector series (13 [↗](#)), and translational fusion was created with the N-terminal as well as C-terminal half of the split YFP fluorophore. NFR1α, SymRKβΔMLD and P2K1 were fused C-terminally to the C-terminal YFP domain. *Nicotiana benthamiana* leaves were infiltrated with *Agrobacterium tumefaciens* GV3101 (grown in LB media supplemented with 50 µg/ml Carbenicillin and 20 µg/ml Gentamicin) carrying the respective constructs and p19 (100 µg/ml Kanamycin and 25 µg/ml Rifampicin) silencing suppressor.

Microscopy was performed 40-46 hours post infiltration. A LEICA SP8 confocal laser scanning microscope with a tunable white light laser (WLL) was used to visualize YFP fluorescence generated upon interaction between the co-expressed proteins as a result of split YFP halves reconstitution. Plasma membrane (PM) was labeled with FM4-64 (Invitrogen, USA) PM dye. YFP was excited at 514 nm and the emission was detected at a 525-575 nm bandwidth. FM4-64 was excited at 510 nm and dye's emission was detected using a 700-780 nm bandwidth. Images of two fluorescence channels were acquired sequentially with a 40x/1.1NA water immersion objective and an additional zoom factor 3. The pixel size of images was set to 95 nm. Brightness and contrast of the images was adjusted in PowerPoint. CLSM was done at the Molecular Cytology Core of the University of Missouri.

Split-Luciferase Complementation Assay

RIN4a/b as well as NFR1α, SymRKβΔMLD and P2K1 were C-terminally fused to split halves of Luciferase in a Split-Luciferase vector system (pCambia-GW-Nluc and pCambia-GW Cluc) and the following constructs were generated via LR reaction: RIN4a/b:Cluc; NFR1α:Nluc, SymRKβΔMLD:Nluc, P2K1:Nluc, and transformed into *Agrobacterium tumefaciens* GV3101 strain. Leaves of 3-weeks old tobacco plants were co-infiltrated (Infiltration buffer: half Murashige and Skoog liquid media and 150 µM Acetosyringone) with agrobacterium carrying the respective constructs at OD600 ~0.55 together with agrobacterium strain carrying p19 silencing suppressor. Protein-protein interaction was observed 2 or 3 dpi, via LUC activity, when Luciferin buffer [100 mM Tris-HCl pH 7.8, 5 mM MgCl₂, 0.15 mM ATP (Sigma), 5 mM D-Luciferin (GoldBio), and 0.01% Silwet-L-77] was sprayed onto the leaves, incubated in dark for 7 min (to decrease autofluorescence originating from chloroplasts) and luminescence was captured by a CCD camera (Photek 216; Photek, Ltd.).

Recombinant protein expression and *in vitro* kinase assay

GmRIN4a, GmRIN4b, GmRIN4c (Glyma.18G166800.1) and GmRIN4d (Glyma.08G349500.1) were fused C-terminally with a His-epitope in the pET22b vector (Novagen). The intracellular, kinase domains of GmSymRK β (537 aa - 919 aa) and NFR1 α (294 aa - 599 aa) were N-terminally tagged with GST-epitope in the pGEX-5X-1 vector (GE Healthcare, Pittsburgh, PA, USA). RosettaTM (DE3) bacterial cells (Novagen/Sigma-Aldrich, Saint Louis, MO, USA) carrying the respective constructs were grown in LB medium (with respective antibiotic) and cell cultures were induced with 0.5 mM isopropyl β -D-1- thiogalactopyranoside (IPTG) after reaching an OD600 absorbance of 0.5 and incubated at 28°C for an additional 4 hours. Bacterial cells were collected at 4,000 rpm for 10 min and His and GST-tagged proteins were purified by TALON Metal Affinity Resin (Clontech, Takara Bio, USA) and Glutathione Resin (GenScript, Piscataway, NJ, USA) following the manufacturer's protocol, respectively. For the *in vitro* kinase assay, 2 μ g of purified GST tagged protein kinases were incubated with 2 μ g His-tagged GmRIN4 proteins as substrate in a 20 μ l reaction buffer containing 20 mM Hepes-KOH (pH 7.4), 5 mM MgCl₂, 1 mM DTT, 2 mM ATP, and w/o 0.2 μ l radioactive [γ -32P] ATP for 60 min at 30°C. In the case of radioactive detection, 5 μ l of 5 $\times$ SDS loading buffer was added to the reaction, and samples were boiled for 5 min. The proteins were separated by electrophoresis in 12% SDS-PAGE gels, followed by autoradiography for 12 h. The proteins were visualized by staining the gel with Coomassie Brilliant Blue (CBB) and auto-radiographed using a Typhoon FLA 9000 phosphorimager (GE Healthcare). Myelin Basic Protein (MBP) and GST were used as controls.

Protein expression and *in planta* phosphorylation in *Arabidopsis* protoplasts

To assess RIN4a and RIN4b phosphorylation *in planta*, the proteins were fused to HA-epitope in a pUGW14 vector driven by 35S promoter and co-expressed with SYMRK Δ MLD (wild type and kinase-dead versions) fused to HA-epitope in *Arabidopsis* leaf protoplasts. Protoplast isolation, transfection and protein extraction from protoplast was performed as described in Cho *et al.*, 2022 (14 [link](#)).

RNA extraction and qRT-PCR experiment

RNA was extracted from entire soybean roots, root hairs and stripped roots, and from transgenic roots using Trizol (Sigma, St. Louis, MO, USA). Tissue was ground in liquid nitrogen, 1 ml 1.5 ml Trizol reagent was added to the mortar and samples were transferred into an Eppendorf tube as liquid. These extracts were incubated on ice for 10 min and then centrifuged at 13000 rcf for 10 min at 4°C. The supernatant was transferred into a new tube and 200 μ l chloroform was added per 1 ml supernatant, vortexed and centrifuged for 20 min at 4°C. The upper phase was carefully transferred into a new tube and the half volume of cold ethanol was added. From this step forward, the samples were transferred onto a Qiagen column, using Qiagen RNeasy extraction kit (Qiagen, Hilden, Germany), following the manufacturer's protocol. The quality of the RNA was checked by agarose-gel electrophoresis and samples were DNase treated using Ambion Turbo DNase (Invitrogen by Fisher Scientific, Vilnius, Lithuania) following the manufacturer's recommendation. cDNA synthesis was performed using 2 μ g RNA, oligo dT primer, and Promega MLV Reverse transcriptase (RT) kit (Madison, WI, USA), a negative control without RT was included in cDNA-synthesis. For qRT-PCR, cDNA was diluted five times and Applied Biosystems SYBR Green (Thermo Fisher Scientific, USA) was used to perform quantitative RT-PCR. For data analysis, Rn values were extracted from ABI 7500 Real Time PCR machine and LinReg software (https://www.gene-quantification.de/LinRegPCR_help_manual_v11.0.pdf [link](#)) was used to determine baseline and Cq values. Data were extracted into Excel file and *Cons6* (F-box protein encoding gene) and/or *Cons7* (Insulin-degrading enzyme, Metalloprotease) was used as a reference gene (15 [link](#)) to normalize Cq values. Δ Ct method (16 [link](#)) was used to evaluate the data and determine relative expression. qPCR primers were designed using Primer3 PCR primer design tool (17 [link](#)). Primers were designed based

on the Williams 82 reference genome. qRT-PCR analysis was performed on cDNA derived from Williams 82 root hairs and stripped roots, and cDNA derived from total root of Bert cultivar and *rin4b* mutant in Bert background.

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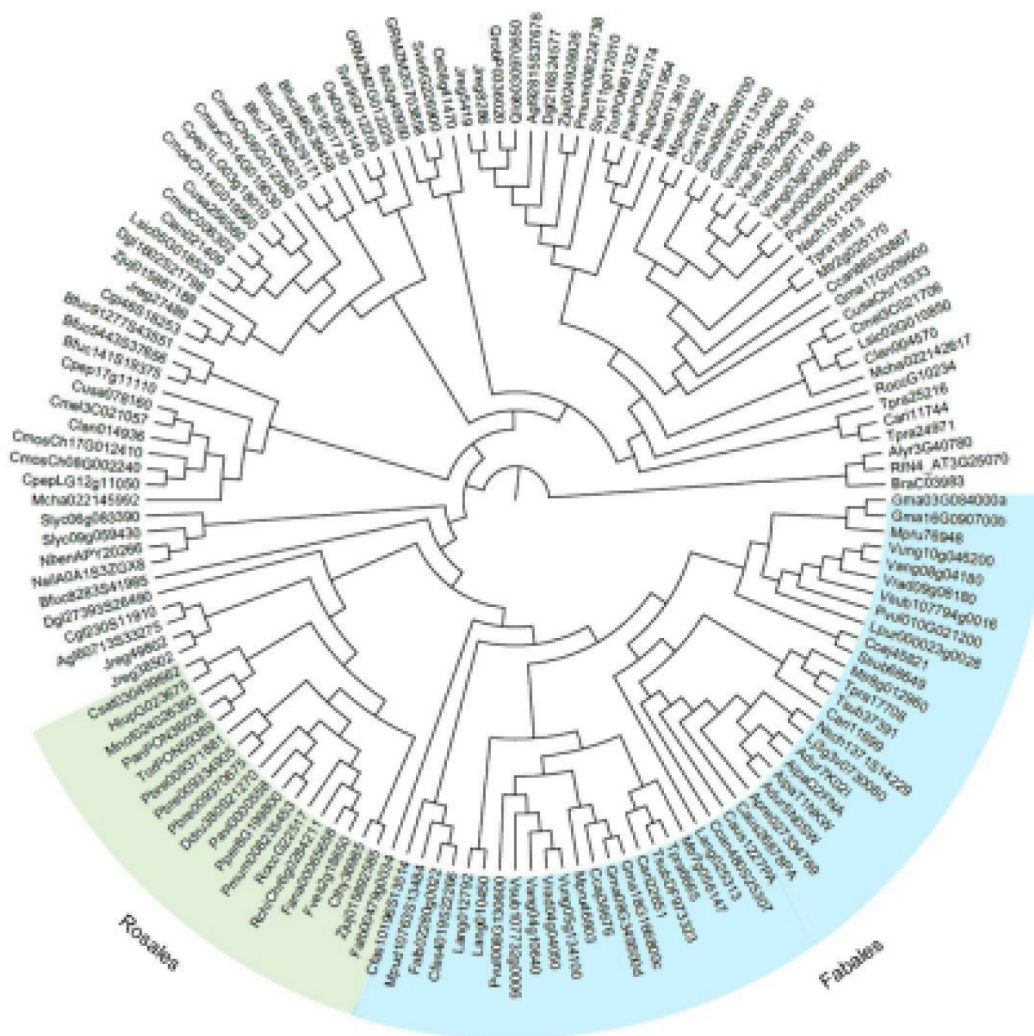


Fig. S1

Phylogenetic tree of RIN4 proteins shows a nodulation-specific subclade

Phylogenetic tree of 149 closest *RIN4* homologs from 66 species, from non-legumes such as *Arabidopsis thaliana*, *A. lyrata*, *Brassica rapa*, *Brachypodium distachyon*, *Setaria viridis*, *Zea mays*, *Oryza sativa*, *Nicotiana benthamiana*, *N. silvestris*, *Solanum lycopersicum*, and from the phylogenetic FaFaCuRo clade: Fabales (nodulating species: *Abrus precatorius*, *Arachis ipaensis*, *A. duranensis*, *Cajanus cajan*, *Chamaecrista fasciculata*, *Cicer arietinum*, *Faidherbia albida*, *Glycine max*, *Lablab purpureus*, *Lotus japonicus*, *Lupinus angustifolius*, *Medicago truncatula*, *Mimosa pudica*, *Mucuna pruriens*, *Phaseolus vulgaris*, *Spatholobus suberectus*, *Trifolium pratense*, *T. subterraneum*, *Vigna radiata*, *V. angularis*, *V. unguiculata*, *V. subterranean*; non-nodulating species: *Chastanosperrum australe*, *Cercis canadensis*, *Nissolia schottii*), Rosales (*Parasponia andersonii* nodulating with rhizobia, its non-nodulating relative *Trema orientale*; *Dryas drummondii* nodulating with Frankia, and non-nodulating *Cannabis sativa*, *Ceanothus thyrsiflorus*, *Fragaria vesca*, *Fragaria x ananassa*, *Humulus lupulus*, *Morus notabilis*, *Prunus persica*, *P. mume*, *P. avium*, *Pyrus bretschneideri*, *Rubus occidentalis*, *Rosa chinensis*, *Ziziphus jujube*), Fagales (nodulating *Alnus glutinosa*, *Casuarina glauca* and non-nodulating *Juglans regia*, *Quercus robur*, *Q. lobata*), and Cucurbitales (nodulating *Datisca glomerata*, non-nodulating *Begonia fuchsioides*, *Citrullus lanatus*, *Cucumis sativus*, *C. pepo*, *C. melo*, *Lagenaria siceraria*, *Momordica charantia*) (37, 52). Two sub-subclades formed: the blue-highlighted contain all the legume *RIN4* homologs, whereas the green contains *RIN4* homologs from Rosales. The tree was build using Average linkage (UPGMA) method and JTT substitution model.

Fig. S2

Alignment of representative RIN4 homologs from the FaFaCuRo clade

The FaFaCuRo clade contains species which are able to form symbiotic nitrogen fixation. Here, we show that the 15 amino acid novel-RIN4-motif (NRM) sequence (red box) and its “GRDSP” core motif (green box) – which is present in soybean (Gma03G084000/RIN4a, Gma16G090700/RIN4b), *L. japonicus* (Lj3g3v0730080), *N. schottii* (Nsch1371S14229) and Parasponia (PanP0836036) RIN4 proteins – is not conserved in nodulating species from Fagales (*Alnus glutinosa* – Agl160713S33275; *Casuarina glauca* – Cgl1230S11910) and Cucurbitales (*Datisca glomerata* – Dgl127393S26490). In addition, non-Fabales species: rice (*Oryza sativa* – Os03g63140), maize (*Zea mays* – ZmB84.05G005000) and poplar (*Populus trichocarpa* – Potri.002G245400) were included in the alignment.



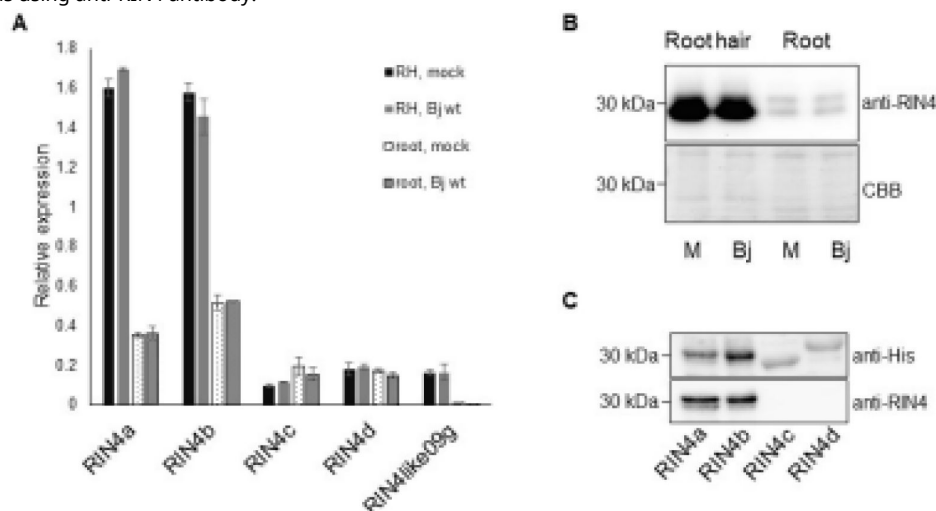
Fig. S3

Soybean RIN4a and RIN4b are highly expressed in root hair and stripped root

(A)

Relative gene expression levels of soybean *RIN4* homologs (*RIN4a-d*) and a *RIN4-like* gene (*Glyma09G008700*). *RIN4a* and *RIN4b* display higher expression levels than *RIN4c*, *RIN4d* and the *RIN4-like* gene in root hair (RH). All four have lower expression levels in roots. No difference could be observed between mock and rhizobial (Bj wt) treatment in root hairs or in roots. qRT-PCR analysis was done on 3 biological replicates, data show the mean of 2 technical replicates of 1 biological replicate.

(B) Immuno-blot analysis, RIN4 proteins detected using RIN4-specific antibody, shows higher protein levels in root hairs than in roots (upper panel). No response was observed to treatment with *B. japonicum* (Bj) in comparison to mock treatment (M). Lower panel: Coomassie Brilliant Blue (CBB) staining of the same membrane showing equal loading. Immuno-blot analysis was performed on three biological replicates. (C) Custom-made anti-RIN4 antibody was tested on recombinantly expressed His-epitope tagged RIN4a, RIN4b, RIN4c and RIN4d proteins. Anti-RIN4 can recognize only RIN4a and RIN4b proteins. Upper panel: Immuno-blot showing RIN4a, b, c, d proteins detected using anti-His antibody. Bottom panel: Immuno-blot detecting RIN4 proteins using anti-RIN4 antibody.



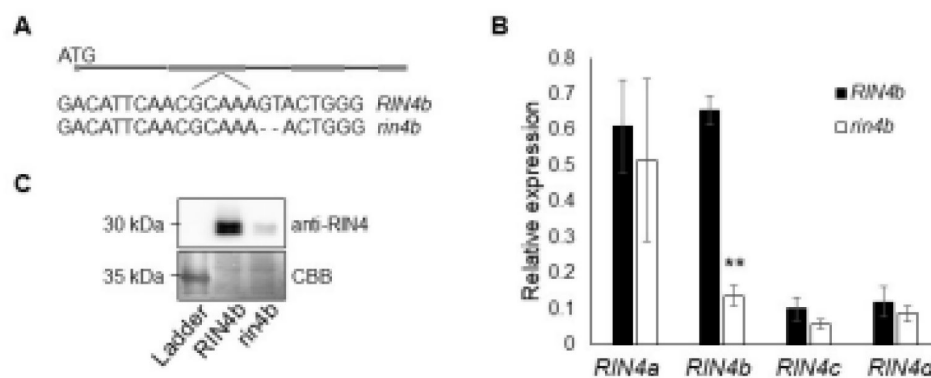


Fig. S4

***RIN4b*-CRISPR-Cas9 deletion and reduced mRNA and protein level in *rin4b*-CRISPR-Cas9 mutant**

(A) Two base pairs deletion was introduced in the second exon of *RIN4b* (in Bert background) using CRISPR-Cas9 technology which led to a pre-mature stop codon in *rin4b* mutant. (B) qRT-PCR analysis performed on roots shows that *RIN4b* mRNA levels were reduced in *rin4b* mutant in comparison to wild-type roots (*RIN4b*), while *RIN4a*, *RIN4c* and *RIN4d* levels were not affected in the *rin4b* mutant background. Error bars represent standard error. Student t-test ** $p < 0.005$. (C) RIN4 protein abundance was reduced in *rin4b*-CRISPR-Cas9 mutant roots (*rin4b*) in comparison to wild type roots (*RIN4b*). Immuno-blot detection was performed using anti-RIN4 antibody on total protein extracted from roots (upper panel). Lower panel: Membrane stained with Coomassie Brilliant Blue (CBB) to show loading.

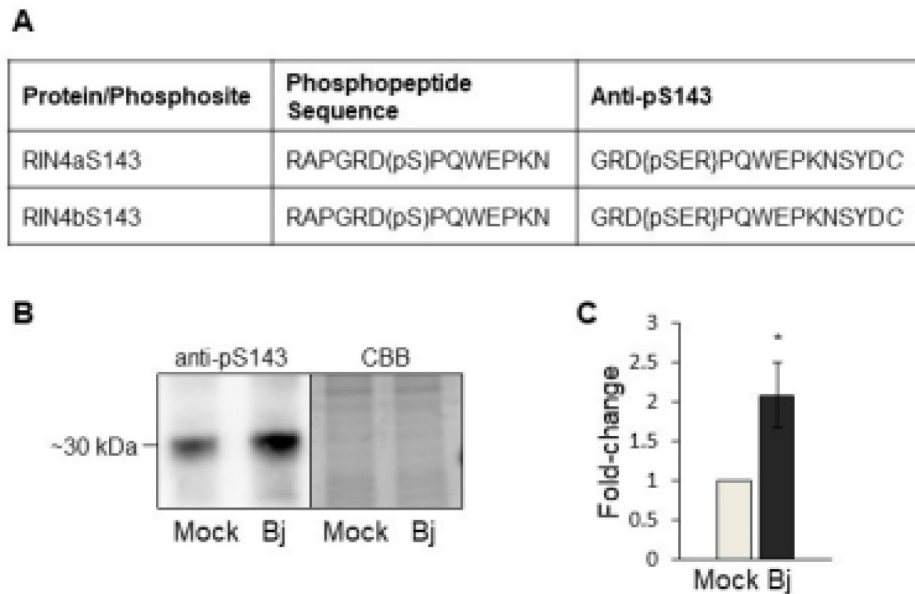


Fig. S5

Nodulation-related S143 phosphorylation site is upregulated in response to rhizobium

(A) Table describing the sequence of the phosphopeptide identified in our previous study ([17](#)). The peptide sequence carrying the S143 phosphorylation sites is identical in RIN4a and RIN4b proteins. The right column shows the sequence of the anti-phosphopeptide against the sequence carrying the S143 phosphorylation site. A cysteine "C" is automatically added to a peptide during synthesis.

(B) Phosphorylation of RIN4^{S143} is up-regulated in soybean root hairs in response to soybean symbiont *B. japonicum*. Left panel: Immune-blot detection using pS143 peptide antibody performed on root hairs treated with H₂O (mock) and soybean symbiont *B. japonicum* (Bj), 1 hpi. Right panel: Coomassie Brilliant Blue (CBB) staining of protein gel previously run to determine equal loading. (C) Quantification of phosphorylation using ImageJ software. Error bar represents standard error. Experiment was repeated three times.

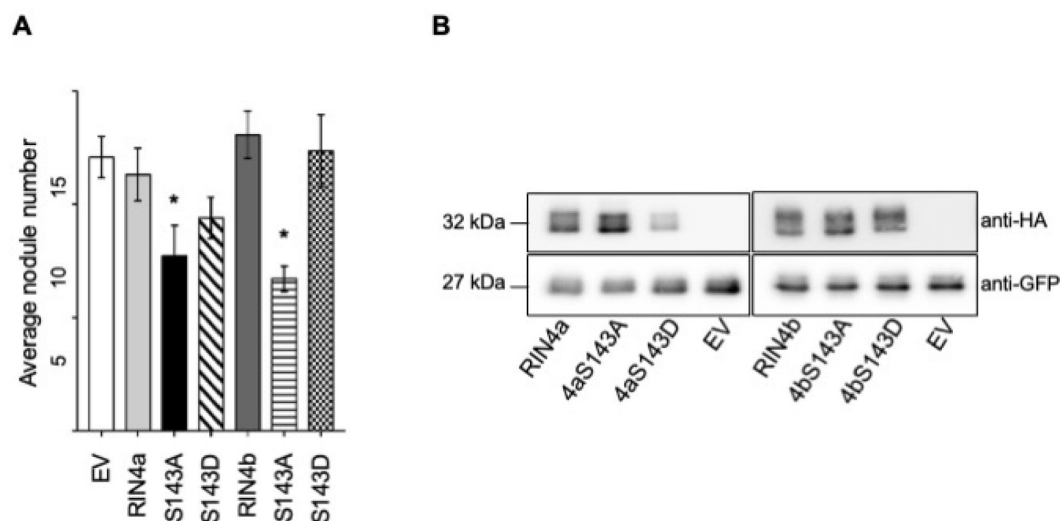


Fig. S6

Novel S143 phosphorylation site contributes to symbiosis development

(A) Phosphor-minus (Ala; S143A) and phosphomimic (Asp; S143D) mutations were introduced into RIN4a and RIN4b to replace S143 residue, and the constructs were expressed in soybean transgenic roots. RIN4a^{S143A} and RIN4b^{S143A} displayed significantly reduced nodule numbers in comparison to transgenic roots carrying empty vector (EV) and the wild-type RIN4a and b protein. RIN4a^{S143D} and RIN4b^{S143D} did not have an effect on nodulation. Roots were phenotyped 5 wpi. Experiment was repeated three times with similar results. Error bars represent standard error. Student's t-test * $p < 0.05$.

(B) Expression of HA-epitope tagged RIN4a and RIN4b and their phosphor-minus (S143A) and phosphor-mimic (S143D) versions in soybean transgenic roots. Immuno-blot analysis confirmed the expression of each version of RIN4a and b. Upper panel: HA antibody detecting RIN4a and b native and mutated proteins. Lower panel: detecting free GFP (27kDa) used as a marker to detect transgenic roots

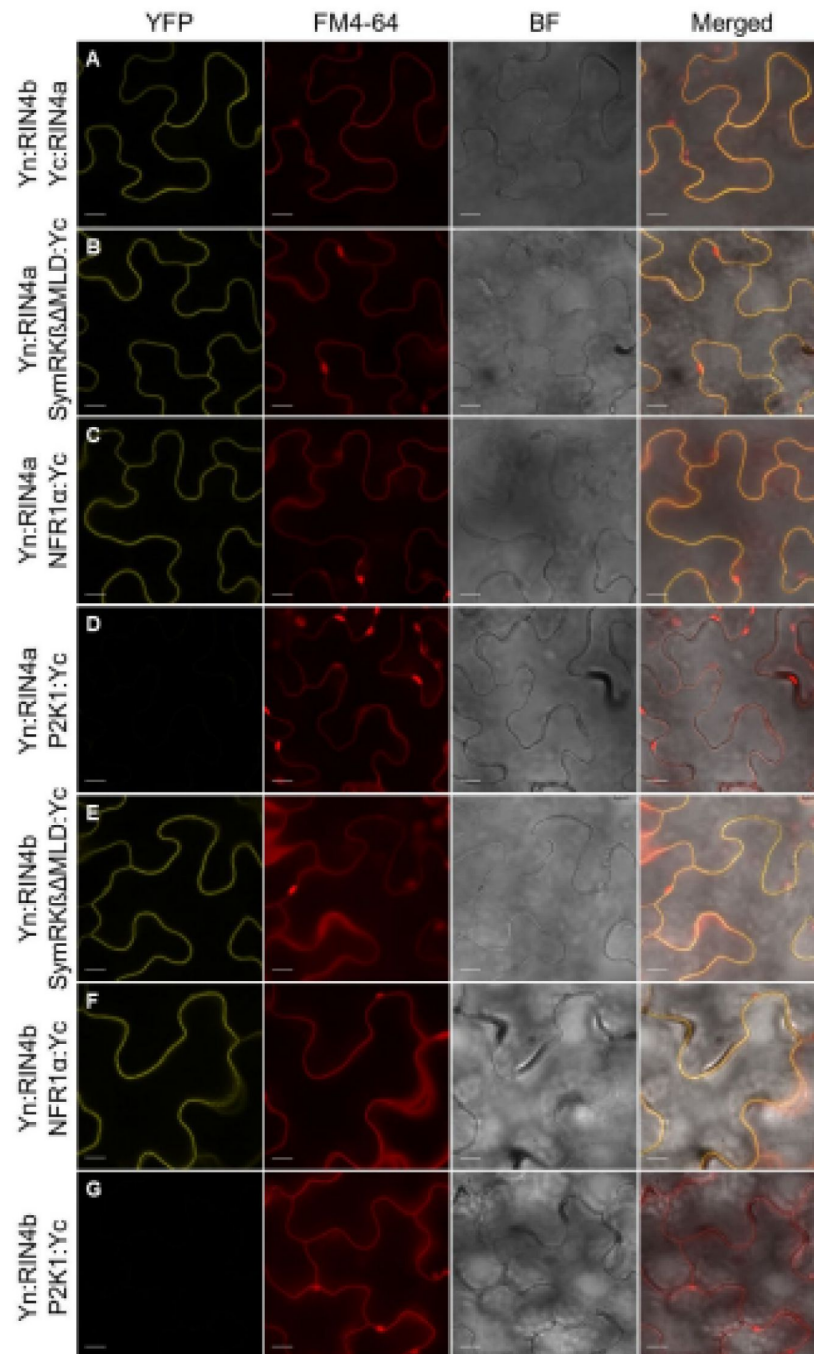


Fig. S8

RIN4a and RIN4b interacts with symbiotic receptor-like kinases NFR1 α and SymRK β in planta

(A) Bimolecular Fluorescence Complementation assay where the RIN4a and RIN4b interaction was used as a positive control. (B) RIN4a interacts with SymRK β Δ MLD (C) and with NFR1 α . (D) No fluorescence signal was observed when RIN4a was co-expressed with the Arabidopsis P2K1 receptor-like kinase, used as a negative control. (E) RIN4b interaction with SymRK β Δ MLD (F) and with NFR1 α . (G) No fluorescence signal was observed when RIN4b was co-expressed with P2K1. Left panels: BiFC, middle left panels FM4-64 staining of the plasma membrane (PM) to show that interaction occurs at the (PM). Middle right panels: bright field (BF). Right panels: merge of YFP (interaction signal) and red PM signal. Scale bars represent 10 μ m. Leica SP8 confocal microscope was used to take images. At least three biological replicates were performed with similar results.

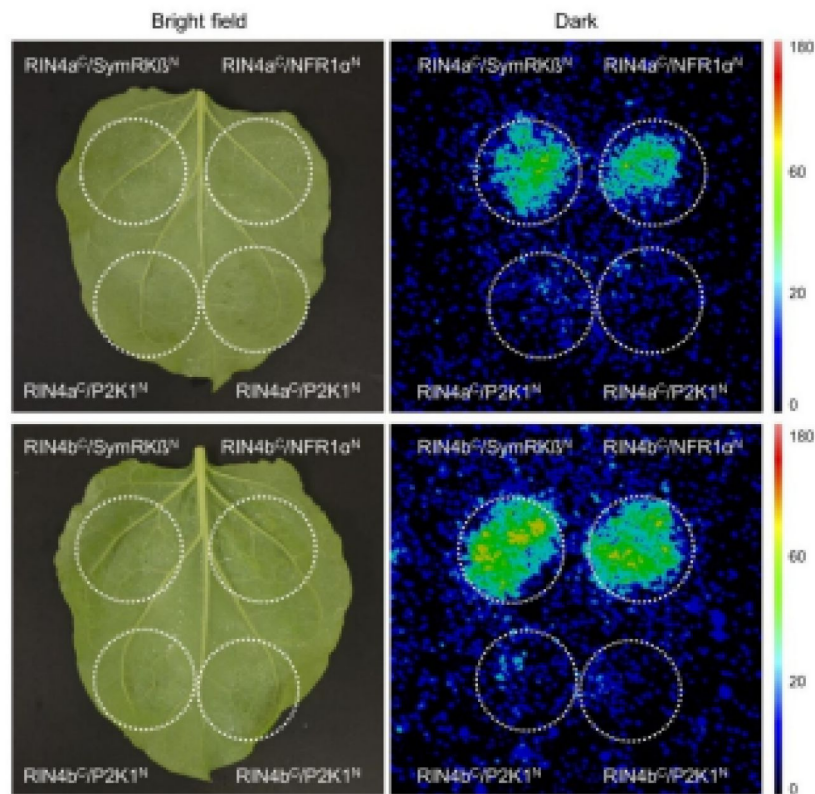


Fig. S9

Split-Luciferase assay confirmed interaction of RIN4a and RIN4b with symbiotic receptors Upper panel shows RIN4a interaction with SymRKβΔMLD and NFR1α.

Upon interaction between the co-expressed proteins fused to split domains of Luciferase, Luciferase activity is restored, and bioluminescence is detected (right panels). Lower panel shows RIN4b interaction with SymRKβΔMLD and NFR1α. In both cases Arabidopsis P2K1 receptor was used as negative control and no bioluminescence could be observed.

Fig. S10

Soybean SymRKβ is an active kinase.

Soybean SymRKβ kinase domain fused to GST was recombinantly expressed and kinase activity was detected *in vitro* using radioactive ATP. SymRKβ-KD trans-phosphorylates MBP substrate (left side). A point mutation was introduced to create a kinase inactive version, SymRKβ^{D734N}, and it resulted in abolished kinase activity (right side, upper panel). Lower panel: CBB staining to show that both native and mutant versions were equally loaded.

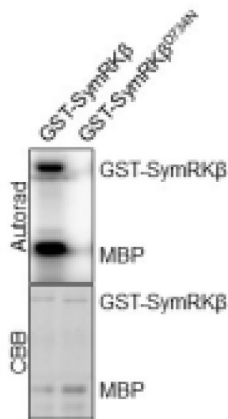
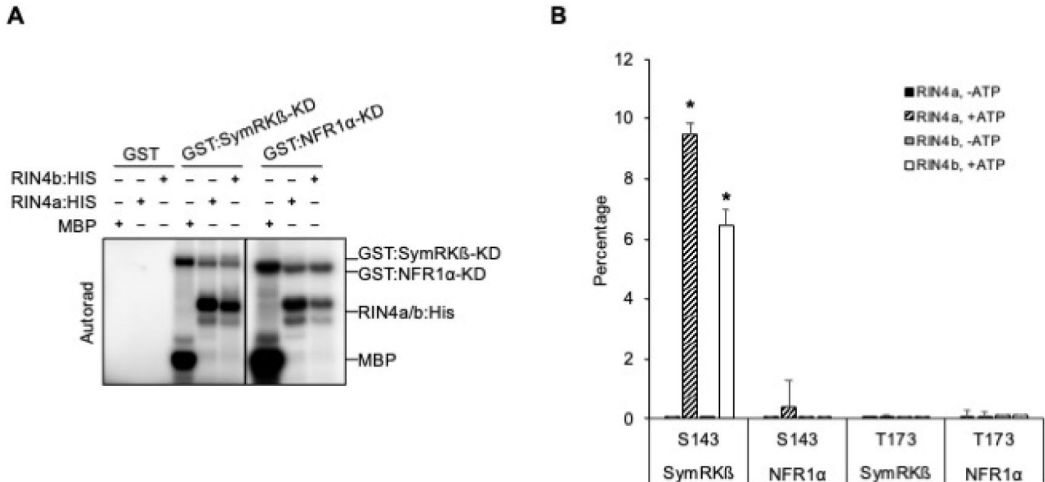


Fig. S11.

In vitro phosphorylation of nodulation-related S143 by SymRKβ.

(A) Recombinantly expressed SymRKβ and NFR1α kinase domains phosphorylate RIN4a and RIN4b *in vitro*. Myelin basic protein (MBP) was used as a positive control for SymRKβ and NFR1α kinase activity, and GST was used (left side) as a negative control. (B) Quantitative mass spectrometry (MS-MRM) was performed to identify the phosphorylation site triggered by the 2 kinases. S143 phosphorylation was phosphorylated only by SymRKβ and quantified using heavy-labeled peptides generated against native peptides carrying S143. In both RIN4a and RIN4b S143 nodulation-related phosphorylation site is phosphorylated by SymRKβ whereas another phosphorylation site T173, used as negative control was not phosphorylated neither by SymRKβ nor by NFR1α. MS run was performed in 2 biological and 3 technical replicates. Phosphorylation is expressed in percentage as the mean of all replicates. Error bars represent standard error. Student's t-test * p<0.05



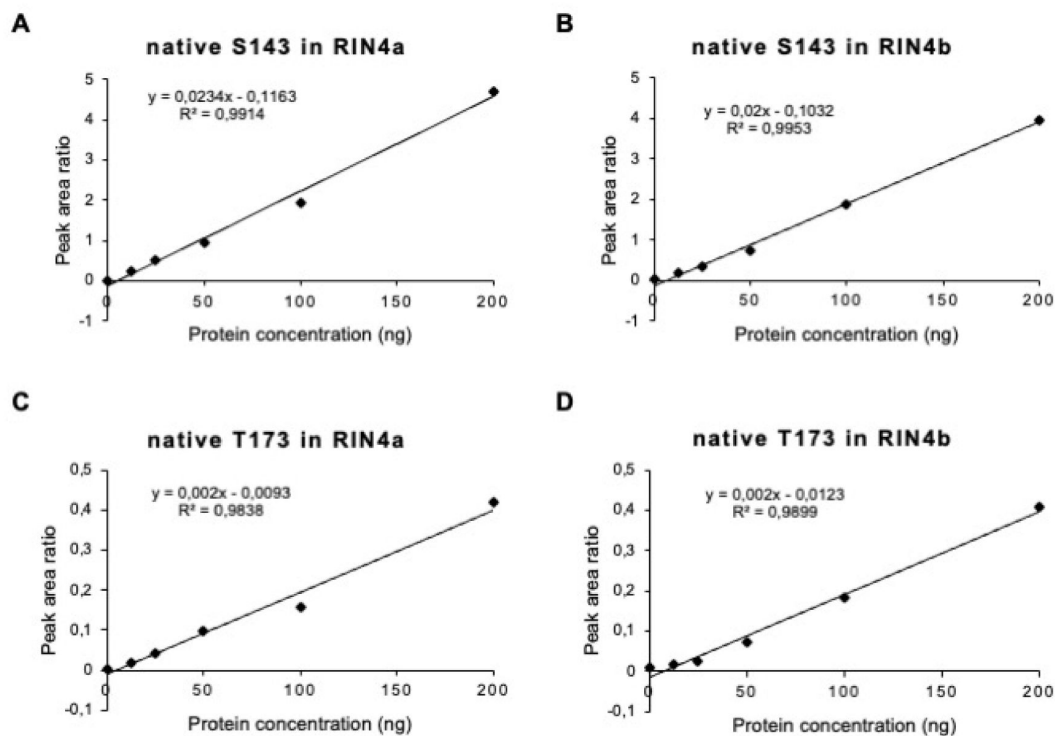


Fig. S12

Linear relationship and correlation coefficients of RIN4a and RIN4b established for MS-SRM Calibration curves of S143 and T173 containing native peptides at different protein concentration of recombinantly expressed RIN4a and RIN4b proteins were established at fixed amount (100 fmol S143; 50 fmol T173) of heavy-labeled AQUA peptide.

Both S143 and T173 carrying peptides are 100% identical between RIN4a and RIN4b.. The experiment was done in RIN4a and RIN4b recombinantly expressed protein background separately. Panels show a linear relationship between peak area ratio of native peptide normalized to heavy-labeled peptide and protein concentration. Scatter plots expressing the relationship between protein concentrations loaded per injection and peak area ratio normalized to heavy obtained per concentration gradient. Correlation coefficient (R^2) values strongly support correlation between the protein concentration injected and the native peptide amount detected (normalized to heavy-labeled peptide).

Full name (Species)	Order		Abbreviation Accession/Transcript ID	Note
<i>Abrus precatorius</i>	Fabales	Apr027334769	XP_027334769	
<i>Arachis ipaensis</i>	Fabales	AipaQ2FNA	Araip.Q2FNA	
		AipaT1NKW	Araip.T1NKW	

Table S1

List of RIN4 proteins used for the phylogenetic tree

<i>A. duranensis</i>	Fabales	Adur7K02I	Aradu.7K02I	
		Adur5N0SW	Aradu.5N0SW	
<i>Cajanus cajan</i>	Fabales	Ccaj45821	Cajcaj_Ccajan_45821	
		Ccaj38676	Cajcaj_Ccajan_38676	
		Ccaj16754	KYP62701	
<i>Chamaecrista fasciculata</i>	Fabales	Cfas4019S22266	Chafas_Chafa4019S22266	
		Cfas10196S13514	Chafas_Chafa10196S13514	
<i>Cicer arietinum</i>	Fabales	Cari11744	Ca_11744	
		Cari22651	Ca_22651	
		Cari11699	Ca_11699	
<i>Faidherbia albida</i>	Fabales	Falb00479g0034	Faialb_Faial00479g0034.1	

		Falb02280g0022	Faialb_Faial02280g0022.1	
<i>Glycine max</i>	Fabales	Gma03G084000a	Glyma.03G084000.1	RIN4a
		Gma16G090700b	Glyma.16G090700.1	RIN4b
		Gma18G166800c	Glyma.18G166800.1	RIN4c
		Gma08G349500d	Glyma.08G349500.1	RIN4d
		Gma09G008700	Glyma.09G008700.1	
		Gma17G009600	Glyma.17G009600.1	
		Gma15G113100	Glyma.15G113100.1	
<i>Lablab purpureus</i>	Fabales	Lpur000023g0028	Labpur_Labpu000023g0028.1	
		Lpur000066g0055	Labpur_Labpu000066g0055.1	
<i>Lotus japonicus</i>	Fabales	Lj3g3v0730080	Lj3g3v0730080.1	

Table S1 (continued)

<i>Lupinus angustifolius</i>	Fabales	Lang025313	OIW18870	
		Lang010450	Lupang_Lup010450.1	
		Lang012792	Lupang_Lup012792.1	
<i>Medicago truncatula</i>	Fabales	Mtr8g012960	Medtr8g012960.2	
		Mtr7g056147	Medtr7g056147.1	
		Mtr2g025170	Medtr2g025170.1	

<i>Mimosa pudica</i>	Fabales	Mpud10763S13484	Mimpud_Mimpu10763S13484	
<i>Mucuna pruriens</i>	Fabales	Mpru76948	Mucpru_lcl_QJKJ01009328.1_RDX76948.1_37901	
		Mpru46803	Mucpru_lcl_QJKJ01012707.1_RDX68051.1_46803	
		Mpru58382	Mucpru_lcl_QJKJ01017556.1_RDX58382.1_55230	
<i>Phaseolus vulgaris</i>	Fabales	Pvul010G021200	Phvul.010G021200.1	
		Pvul008G130600	Phvul.008G130600.1	
		Pvul006G144600	Phvul.006G144600.1	
<i>Spatholobus suberectus</i>	Fabales	Ssub68649	TKY68649	
<i>Trifolium pratense</i>	Fabales	Tpra17708	Tp57577_TGAC_v2_mRNA17708	
		Tpra6665	Tp57577_TGAC_v2_mRNA6665	
		Tpra13613	Tp57577_TGAC_v2_mRNA13613	
		Tpra25216	Tp57577_TGAC_v2_mRNA25216	
		Tpra24971	Tp57577_TGAC_v2_mRNA24971	
<i>T. subterraneum</i>	Fabales	Tsub37391	Trisub_lcl_DF974408.1_GAU48333.1_37391	

Table S1 (continued)

		TsubDF973323	Trisub_lcl_DF973323.1_GAU25883.1_14941	
<i>Vigna angularis</i>	Fabales	Vang08g04180	XP_017442542	
		Vang04g10640	Vigang_vigan.Vang04g10640.1	
		Vang03g07180	Vigang_vigan.Vang03g07180.1	
<i>Vigna radiata</i>	Fabales	Vrad09g08180	A0A1S3VF94	
		Vrad04g04050	Vigrad_Vradi04g04050.1	
		Vrad10g07710	Vigrad_Vradi10g07710.1	
<i>V. subterranean</i>	Fabales	Vsub107794g0016	Vigsub_Vigsu107794g0016.1	
		Vsub107732g0006	Vigsub_Vigsu107732g0006.1	
		Vsub107920g0110	Vigsub_Vigsu107920g0110.1	
<i>V. unguiculata</i>	Fabales	Vung10g046200	Vigun10g046200.1	
		Vung05g134100	Vigun05g134100.1	
		Vung06g156400	Vigun06g156400.1	
<i>Chastanospermum australe</i>	Fabales	Caus1227PA	Caus_Castanospermum21227-PA	
		Caus26878PA	Caus_Castanospermum26878-PA	
<i>Cercis canadensis</i>	Fabales	Ccan480S25307	Cercan_Cerca480S25307	
		Ccan86S33867	Cercan_Cerca86S33867	
<i>Nissolia schottii</i>	Fabales	Nsch1371S14229	Nissch_Nissc1371S14229	
		Nsch15112S15091	Nissch_Nissc15112S15091	
<i>Parasponia andersonii</i>	Rosales	PanPON36036	PON36036	
		PanPON52174	PON52174	

Table S1 (continued)

<i>Trema orientale</i>	Rosales	TorPON59389	PON59389	
		TorPON91322	PON91322	
<i>Dryas drummondii</i>	Rosales	Ddru380S21270	Drydru_Drydr380S21270	
<i>Cannabis sativa</i>	Rosales	Csat030499662	XP_030499662	
<i>Ceanothus thyrsiflorus</i>	Rosales	Cthy36985	c36985_g1_i2	
<i>Fragaria vesca</i>	Rosales	Fves2g18650	Fraves_FvH4_2g18650.1	
<i>Fragaria x ananassa</i>	Rosales	Fana00065658	Fraana_FAN_iscf00065658.1.g00001.1	
<i>Humulus lupulus</i>	Rosales	HlupG023675	Humlup_HL.SW.v1.0.G023675.1	
		HlupG031954	Humlup_HL.SW.v1.0.G031954.1	
<i>Morus notabilis</i>	Rosales	Mnot024026395	XP_024026395	
		Mnot013610	EXB41536	
<i>Prunus persica</i>	Rosales	Pper8G199800	Pruper_Prupe.8G199800.1	
<i>P. mume</i>	Rosales	Pmum008235453	Prumum_lcl_NC_024131.1_XP_008235453.1_19524	
		Pmum008224738	Prumum_lcl_NC_024128.1_XP_008224738.1_9378	
<i>P. avium</i>	Rosales	Pavi0000558	Pruavi_Pav_sc0000558.1_g920.1.mk	
<i>Pyrus bretschneideri</i>	Rosales	Pbre009370675	Pyrbre_lcl_NW_008988223.1_XP_009370675.1_22569	
		Pbre009334905	Pyrbre_lcl_NW_008988430.1_XP_009334905.1_32138	
		Pbre009371881	Pyrbre_lcl_NW_008988045.1_XP_009371881.1_2472	

Table S1 (continued)

<i>Rubus occidentalis</i>	Rosales	RoccG22517	Rubocc_Bras_G22517	
		RoccG10234	Rubocc_Bras_G10234	
<i>Rosa chinensis</i>	Rosales	RchiChr6g0284211	A0A2P6PU72	
<i>Ziziphus jujube</i>	Rosales	Zjuj015892585	Zizjuj_lcl_NC_029687.1_XP_015892585.1_24446	
		Zjuj015887188	Zizjuj_lcl_NC_029685.1_XP_015887188.1_19239	
		Zjuj024926926	Zizjuj_lcl_NC_029679.1_XP_024926926.1_1025	
<i>Alnus glutinosa</i>	Fagales	Agl60713S33275	Alngl60713S33275	
		Agl92815S37678	Alngl92815S37678	
<i>Casuarina glauca</i>	Fagales	Cgl230S11910	Casgl230S11910	
		Cgl46S18253	Casgl46S18253	
<i>Juglans regia</i>	Fagales	Jreg38502	Jugreg_g38502.t1	
		Jreg49802	Jugreg_g49802.t1	
		Jreg27489	Jugreg_g27489.t1	
		Jreg5419	Jugreg_g5419.t1	
		Jreg8286	Jugreg_g8286.t1	
<i>Quercus robur</i>	Fagales	QrobP0336020	Querob_Qrob_P0336020.2	

<i>Q. lobata</i>	Fagales	Qlob030970650	XP_030970650	
<i>Datisca glomerata</i>	Cucurbitales	Dgl27393S26490	Datgl27393S26490	
		Dgl216S24577	Datgl216S24577	
		Dgl1602S21798	Datgl1602S21798	
<i>Begonia fuchsoides</i>	Cucurbitales	Bfuc8283S41985	Begfuc_Begfu8283S41985	

Table S1 (continued)

		Bfuc141S19375	Begfuc_Begfu141S19375	
		Bfuc5443S37656	Begfuc_Begfu5443S37656	
		Bfuc91277S43551	Begfuc_Begfu91277S43551	
		Bfuc719S40210	Begfuc_Begfu719S40210	
		Bfuc646S39458	Begfuc_Begfu646S39458	
		Bfuc2878S29171	Begfuc_Begfu2878S29171	
<i>Citrullus lanatus</i>	Cucurbitales	Clan014936	Citlan_Cla014936	
		Clan021409	Citlan_Cla021409	
		Clan004570	Citlan_Cla004570	
<i>Cucumis sativus</i>	Cucurbitales	Cusa078160	Cucsa.078160.1	
		Cusa256560	Cucsa.256560.2	
		CusaChr13333	XP_011660296	

<i>Cucumis melo</i>	Cucurbitales	Cmel3C021057	XP_008458026	
		CmelC006302	XP_008438343	
		Cmel3C021706	Cucmel_MELO3C021706.2.1	
<i>Cucurbita pepo</i>	Cucurbitales	CpepLG12g11050	Cucpep_Cp4.1LG12g11050.1	
		Cpep17g11110	Cucpep_Cp4.1LG17g11110.1	
		Cpep1LG03g18010	Cucpep_Cp4.1LG03g18010.1	
<i>Cucurbita maxima</i>		CmaxCh14G019030	Cucmax_CmaCh14G019030.1	
		CmaxCh06G012380	Cucmax_CmaCh06G012380.1	
<i>Cucurbita moschata</i>		CmosCh08G002240	XP_022953250	
		CmosCh17G012410	Cucmos_CmoCh17G012410.1	
		CmosCh14G019560	Cucmos_CmoCh14G019560.1	

Table S1 (continued)

<i>Lagenaria siceraria</i>	Cucurbitales	Lsic05G018530	Lagsic_Lsi05G018530.1	
		Lsic02G010850	Lagsic_Lsi02G010850.1	
<i>Momordica charantia</i>	Cucurbitales	Mcha022145992	XP_022145992	
		Mcha022142617	Momcha_lcl_NW_019104509.1_XP_022142617.1_13621	

	Non FaFaCu Ro			
<i>Arabidopsis thaliana</i>	Non FaFaCu Ro	RIN4 AT3G25070	AT3G25070	AtRIN4
<i>A. lyrata</i>	Non FaFaCu Ro	Alyr3G40780	AL3G40780.t1	
<i>Brassica rapa</i>	Non FaFaCu Ro	BraC03983	Brara.C03983.1	
<i>Brachypodium distachyon</i>	Non FaFaCu Ro	Bdi1g01730	Bradi1g01730.3	
		Bdi3g40950	Bradi3g40950.1	
<i>Setaria viridis</i>	Non FaFaCu Ro	Svir6G226900	Sevir.6G226900.1	
		Svir9G012200	Sevir.9G012200.1	
<i>Zea mays</i>	Non FaFaCu Ro	GRMZM2G012229	GRMZM2G012229_T01	
		GRMZM2G703858	GRMZM2G703858_T02	
<i>Oryza sativa</i>	Non FaFaCu Ro	Os03g63140	LOC_Os03g63140.1	
		Os08g41470	LOC_Os08g41470.1	

Table S1 (continued)

Table S1 (continued)

<i>Nicotiana benthamiana</i>	Non FaFaCu Ro	NbenAPY20266	APY20266	
<i>N. silvestris</i>	Non FaFaCu Ro	NsilA0A1S3ZGX8	A0A1U7YR47	
<i>Solanum lycopersicum</i>	Non FaFaCu Ro	Slyc09g059430	Solyc09g059430.2.1	
		Slyc06g083390	Solyc06g083390.2.1	
		Slyc11g012010	Solyc11g012010.2.1	

Table S2**Summary of *L. japonicus rin4* mutant screen**

Mutant ID	Total No. of plants	Wild type	Heterozygous	Homozygous
30019656	8	3	5	0
30000711	10	3	7	0

Peptide name	Native peptide sequence	Heavy-labeled peptide sequence	Note
S143 (non phosphorylated)	APGRDSPQWEPK	R.APGRDSPQWEP.[K_C13N15]	RIN4a and RIN4b sequences are identical
pS143 (phosphorylated)	APGRD(pS)PQWEPK	R.APGRD(pS)PQWEP.[K_C13N15]	RIN4a and RIN4b sequences are identical
T173 (non phosphorylated)	GDETPDKGAAV PK	R.GDETPDKGAAVP.[K_C13N15]	RIN4a and RIN4b sequences are identical
pT173 (phosphorylated)	GDE(pT)PDKGAAV PK	R.GDE(pT)PDKGAAVP.[K_C13N15]	RIN4a and RIN4b sequences are identical

Table S3

Soybean RIN4 native and AQUA peptides

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Author information

Katalin Tóth[†]

Division of Plant Science and Technology, C.S. Bond Life Sciences Center, University of Missouri, Columbia, United States

For correspondence: katalintoht2005@gmail.com

[†]Present address: Inari Agriculture NV, Ghent, Belgium

Daewon Kim

Division of Plant Science and Technology, C.S. Bond Life Sciences Center, University of Missouri, Columbia, United States

Sung-Hwan Cho

Division of Plant Science and Technology, C.S. Bond Life Sciences Center, University of Missouri, Columbia, United States

Cuong T Nguyen

Cuu Long Delta Rice Research Institute, Cần Thơ City, Vietnam

Tran HN Nguyen

Department of Molecular and Environmental Biotechnology, University of Science, VNU HCMC, Ho Chi Minh City, Vietnam

Christopher Hartanto

Division of Biochemistry, University of Missouri, Columbia, United States

Jean-Michel Michno

Department of Agronomy and Plant Genetics, University of Minnesota, St. Paul, United States

Adrian O Stec

Department of Agronomy and Plant Genetics, University of Minnesota, St. Paul, United States

Robert M Stupar

Department of Agronomy and Plant Genetics, University of Minnesota, St. Paul, United States

Gary Stacey

Division of Plant Science and Technology, C.S. Bond Life Sciences Center, University of Missouri, Columbia, United States, Division of Biochemistry, University of Missouri, Columbia, United States

ORCID iD: [0000-0001-5914-2247](https://orcid.org/0000-0001-5914-2247)

For correspondence: staceyg@missouri.edu

Editors

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Reviewer #1 (Public review):

The authors set out to illuminate how legumes promote symbiosis with beneficial nitrogen fixing bacteria while maintaining a general defensive posture towards the plethora of potentially pathogenic microbes in their environment. Intriguingly, a protein involved in plant defence signalling, RIN4, is implicated as a type of 'gatekeeper' for the symbiosis, connecting symbiosis signalling with defence signalling. Although questions remain about how exactly RIN4 enables the symbiosis, the work opens an important door to new discoveries in this area.

Strengths:

The study uses a multidisciplinary, state-of-the-art approach to implicate RIN4 in soybean nodulation and symbiosis development. The results support the authors' conclusions.

Weaknesses:

None after thoughtful revision.

<https://doi.org/10.7554/eLife.93149.2.sa2>

Reviewer #3 (Public review):

Summary:

This manuscript by Toth et al reveals a conserved phosphorylation site within the RIN4 (RPM1-interacting protein 4) R protein that is exclusive to two of the four nodulating clades, Fabales and Rosales. The authors present persuasive genetic and biochemical evidence that phosphorylation at the serine residue 143 of GmRIN4b, located within a 15-aa conserved motif with a core five amino acids 'GRDSP' region, by SymRK, is essential for optimal nodulation in soybean. The experimental design and results are robust, the manuscript's discussion has been satisfactorily updated. Results described here are important to understand how the symbiosis signaling pathway prioritizes associations with beneficial rhizobia, while repressing immunity-related signals.

Strengths:

The manuscript asks an important question in plant-microbe interaction studies with interesting findings.

Overall, the experiments are detailed, thorough and very well-designed. The findings appear to be robust.

The authors provide results that are not overinterpreted and are instead measured and logical.

Weaknesses:

No major weaknesses.

<https://doi.org/10.7554/eLife.93149.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

The authors set out to illuminate how legumes promote symbiosis with beneficial nitrogen-fixing bacteria while maintaining a general defensive posture towards the plethora of potentially pathogenic bacteria in their environment. Intriguingly, a protein involved in plant defence signalling, RIN4, is implicated as a type of 'gatekeeper' for symbiosis, connecting symbiosis signalling with defence signalling. Although questions remain about how exactly RIN4 enables symbiosis, the work opens an important door to new discoveries in this area.

Strengths:

The study uses a multidisciplinary, state-of-the-art approach to implicate RIN4 in soybean nodulation and symbiosis development. The results support the authors' conclusions.

Weaknesses:

No serious weaknesses, although the manuscript could be improved slightly from technical and communication standpoints.

Reviewer #2 (Public Review):

Summary:

The study by Toth et al. investigates the role of RIN4, a key immune regulator, in the symbiotic nitrogen fixation process between soybean and rhizobium. The authors found that SymRK can interact with and phosphorylate GmRIN4. This phosphorylation occurs within a 15 amino acid motif that is highly conserved in Nfixation clades. Genetic studies indicate that GmRIN4a/b play a role in root nodule symbiosis. Based on their data, the authors suggest that RIN4 may function as a key regulator connecting symbiotic and immune signaling pathways.

Overall, the conclusions of this paper are well supported by the data, although there are a few areas that need clarification.

Strengths:

This study provides important insights by demonstrating that RIN4, a key immune regulator, is also required for symbiotic nitrogen fixation.

The findings suggest that GmRIN4a/b could mediate appropriate responses during infection, whether it is by friendly or hostile organisms.

Weaknesses:

The study did not explore the immune response in the rin4 mutant. Therefore, it remains unknown how GmRIN4a/b distinguishes between friend and foe.

Reviewer #3 (Public Review):

Summary:

This manuscript by Toth et al reveals a conserved phosphorylation site within the RIN4 (RPM1-interacting protein 4) R protein that is exclusive to two of the four nodulating clades, Fabales and Rosales. The authors present persuasive genetic and biochemical evidence that phosphorylation at the serine residue 143 of GmRIN4b, located within a 15-aa conserved motif with a core five amino acids 'GRDSP' region, by SymRK, is essential for optimal nodulation in soybean. While the experimental design and results are robust, the manuscript's discussion fails to clearly articulate the significance of these findings. Results described here are important to understand how the symbiosis signaling pathway prioritizes associations with beneficial rhizobia, while repressing immunity-related signals.

Strengths:

The manuscript asks an important question in plant-microbe interaction studies with interesting findings.

Overall, the experiments are detailed, thorough, and very well-designed. The findings appear to be robust.

The authors provide results that are not overinterpreted and are instead measured and logical.

Weaknesses:

No major weaknesses. However, a well-thought-out discussion integrating all the findings and interpreting them is lacking; in its current form, the discussion lacks 'boldness'. The primary question of the study - how plants differentiate between pathogens and symbionts - is not discussed in light of the findings. The concluding remark, "Taken together, our results indicate that successful development of the root nodule symbiosis requires cross-talk between NF-triggered symbiotic signaling and plant immune signaling mediated by RIN4," though accurate, fails to capture the novelty or significance of the findings, and left me wondering how this adds to what is already known. A clear conclusion, for eg, the phosphorylation of RIN4 isoforms by SYMRK at S143 modulates immune responses during symbiotic interactions with rhizobia, or similar, is needed.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

I have no major criticism of the work, although it could be improved by addressing the following minor points:

(1) Page 8, Figure 2 legend. Consider changing "proper symbiosis formation" to "normal nodulation" or something that better reflects control of nodule development/number.

We thank you for the suggestion, the legend was changed to “...required for normal nodule formation” (see Page 10, revised manuscript)

(2) Page 9. Cut "newly" from the first sentence of paragraph 2, as S143 phosphorylation was identified previously.

Thank you for the suggestion, we removed “newly” from the sentence.

(3) Page 10, Figure 3. Panels B showing green-fluorescent nodules are unnecessary given the quantitative data presented in the accompanying panel A. This goes for similar supplemental figures later.

We appreciate the comment; regarding Figure 3 (complementing *rin4b* mutant, we updated the figures according to the other reviewer’s comment) and Suppl Figure 6 (OE phenotype of phospho-mimic/negative mutants), we removed the panels showing the micrographs. At the same time, we did not modify Figure 2 (where micrographs showing transgenic roots carrying the silencing constructs) for the sake of figure completeness. (See Page 10, revised manuscript)

(4) Consider swapping Figure 3 for Supplemental Figure S7, which I think shows more clearly the importance of RIN4 phosphorylation in nodulation.

We appreciate the comment and have swapped the figures according to the reviewer’s suggestion. Legend, figure description, and manuscript text have been updated accordingly. (See page 12 and 38, revised manuscript)

(5) Page 10. Replace "it will be referred to S143..." with "we refer to S143 instead of".

We replaced it according to the comment.

(6) Page 11, delete "While" from "While no interactions could be observed..."

We deleted it according to the suggestion.

(7) Page 33, Fig S5. How many biological replicates were performed to produce the data presented in panel C and what do the error bar and asterisk indicate? Check that this information is provided in all figures that show errors and statistical significance.

Thank you for the remark. The experiment was repeated three times, and this note was added to the figure description. All the other figure legends with error bar(s) were checked whether replicates are indicated accordingly.

(8) Page 37, Fig S11, panel B. Are averages of data from the 2 biological and 3 technical replicates shown? Add error bars and tests of significant difference.

Averages of a total of 6 replicates (from 2 biological replicates, each run in triplicates) are shown. We thank the reviewer for pointing out the missing error bars and statistical test, we have updated the figure accordingly.

(9) Fig S12. Why are panels A, C, E, and G presented? The other panels seem to show the same data more clearly- showing the linear relationship between peak area ratio and protein concentration.

We have taken the reviewer's comment into consideration and revised the figure, removing the calibration curves and showing only four panels. The figure legend has been corrected accordingly. (Please see page 43, revised manuscript). The original figure (unlike other revised figures) had to be deleted from the revised manuscript, as it caused technical issues when converting the document into pdf.

Reviewer #2 (Recommendations For The Authors):

Some small suggestions:

(1) It's good to include a protein schematic for RIN4 in Figure 1.

We appreciate the reviewer's suggestion and we have drawn a protein schematic and added it to Figure 1. The figure legend was updated accordingly.

(2) There appears to be incorrect labeling in Figure 2c; please double-check and make the necessary corrections.

With respect, we do not understand the comment about incorrect labeling. Would the reviewer please help us out and give more explanation? In Figure 2C, RIN4a and RIN4b expression was checked in transgenic roots expressing either EV (empty vector) or different silencing constructs targeting RIN4a/b.

Reviewer #3 (Recommendations For The Authors):

I enjoyed the level of detail and precision in experimental design.

A discussion point could be - What does it mean that nodule number but not fixation is affected? Is RIN4 only involved in the entry stage of infection but not in nodules during N-fixation?

Current/Our data suggest that RIN4 does indeed appear to be involved in infection. This hypothesis is supported by the findings that RIN4a/b was found phosphorylated in root hairs

but not in root (or it was not detected in the root). The interaction with the early signaling RLKs also suggests that RIN4 is likely involved in the early stage of symbiosis formation.

How would the authors explain their observation "However, the motif is retained in non-nodulating Fabales (such as C. canadensis, N. schottii; SI Appendix, Figure S2) and Rosales species as well." What does this imply about the role in symbiosis that the authors propose?

We appreciate the reviewer's question. The motif seems to be retained, however, it might be not only the motif but also the protein structure that in case of nodulating plants might be different. We have not investigated the structure of RIN4, how it would look based on certain features/upon interaction with another protein and/or post-translational modification(s). Griesman et al, (2018) showed the absence of certain genes within Fabales in non-nodulating species, we can speculate that these absent genes can't interact with RIN4 in those species, therefore the lack of downstream signaling could be possible (in spite of the retained motif in non-nodulating species). At this point, there is not enough data or knowledge to further speculate.

qPCR analysis of symbiotic pathway genes showed that both NIN-dependent and NIN-independent branches of the symbiosis signaling pathway were negatively affected in the rin4b mutant. Please derive a conclusion from this.

We appreciate the comment, it also prompted us to correct the following sentence; original: "Since NIN is responsible for induction of NF-YA and ERN1 transcription factors, their reduced expression in rin4b plants was not unexpected (Fig. 5). "As ERN1 expression is independent of NIN (Kawaharada et al, 2017). The following sentences were also deleted as it represented a repetition of a statement above these sentences: "Soybean NF-YA1 homolog responded significantly to rhizobial treatment in rin4b plants, whereas NF-YA3 induction did not show significant induction (Fig. 5)."

We added the following conclusion/hypothesis: "Based on the results of the expression data presented above, it seems that both NIN-dependent and NINindependent branches of the symbiotic signaling pathways are affected in the rin4b mutant background. This indicates that the role of RIN4 protein in the symbiotic pathway can be placed upstream of CYCLOPS, as the CYCLOPS transcription activating complex is responsible (directly or indirectly) for the activation of all TFs tested in our expression analysis (Singh et al, 2014/47, 48)." (Please see Page 16, revised manuscript)

The authors are highly encouraged to write a thoughtful discussion that would accompany the detailed experimental work performed in this manuscript.

We appreciate the comment, and we did some work on the discussion part of the document. (Please see Pages 17-19, revised manuscript)

Some minor suggestions for overall readability are below.

What about immune signaling genes? Given that authors hypothesize that "Absence of AtRIN4 leads to increased PTI responses and, therefore, it might be that GmRIN4b absence also causes enhanced PTI which might have contributed to significantly fewer nodules." Could check marker immune signaling gene expression FLS2 and others.

We appreciate the reviewer's comment, and while we believe those are very interesting questions/suggestions, answering them is out of the scope of the current manuscript. Partially because it has been shown that several defenseresponsive genes that were described in leaf immune responses could not be confirmed to respond in a similar manner in root (Chuberre

et al., 2018). It was also shown that plant immune responses are compartmentalized and specialized in roots (Chubierre et al., 2018). If we were looking at immune-responsive genes, the signal might be diluted because of its specialized and compartmentalized nature. Another reason why these questions cannot be answered as a part of the current manuscript is because finding a suitable immune responsive gene would require rigorous experiments (not only in root, but also in root hair (over a timecourse) which would be a ground work for a separate study (root hair isolation is not a trivial experiment, it requires at least 250-300 seedlings per treatment/per time-point).

Regarding FLS2, it is known in Arabidopsis that its expression is tissue-specific within the root, and it seems that FLS2 expression is restricted to the root vasculature (Wyrsh et al, 2015). In our manuscript, we showed that RIN4a/b is highly expressed in root hairs, as well as RIN4 phosphorylation was detectable in root hair but not in the root; therefore, we do not see the reason to investigate FLS2 expression.

"in our hands only ERN1a could be amplified. One possible explanation for this observation is that primers were designed based on Williams 82 reference genome, while our rin4b mutant was generated in the Bert cultivar background." Is the sequence between the two cultivars and the primers that bind to ERN1b in both cultivars so different? If not, this explanation is not very convincing.

At the time of performing the experiment the genomic sequence of the Bert cultivar (used for generating rin4b edited lines) was not publicly available. In accordance with the reviewer's comment, we removed the explanation, as it does not seem to be relevant. (See page 16, revised manuscript)

The figures are clear and there is a logical flow. The images of fluorescing nodules in Figure 2,3 panels with nodules are not informative or unbiased .

We appreciate the comment, as for Figure 3 (complementing rin4b mutant), we updated the figures according to the other reviewer's comment and Suppl. Figure 6 (OE phenotype of phospho-mimic/negative mutants) we removed the panels showing the micrographs. At the same time, we did not modify Figure 2 (where micrographs showing transgenic roots carrying the silencing constructs) for the sake of figure completeness. (See pages 10, 12 and 38, revised manuscript)

What does the exercise in isolation of rin4 mutants in lotus tell us? Is it worth including?

Isolation of the Ljrin4 mutant suggests that RIN4 carries such an importance that the mutant version of it is lethal for the plant (as in Arabidopsis, where most of the evidence regarding the role of RIN4 has been described), and an additional piece of evidence that RIN4 is similarly crucial across most land plant species.

Sentence ambiguous. "Co-expression of RIN4a and b with SymRKβΔMLD and NFR1a _resulted in YFP fluorescence detected by Confocal Laser Scanning Microscopy (SI Appendix, Figure S8) suggesting that RIN4a and b proteins closely associate with both RLKs." Were all 4 expressed together?

Thank you for the remark. Not all 4 proteins were co-expressed together. We adjusted the sentence as follows: "Co-expression of RIN4a/ and b with SymRKβΔMLD as well as and NFR1a resulted in YFP fluorescence..." I hope it is phrased in a clearer way. (See page 13, revised manuscript)

Minor spelling errors throughout.. Costume-made (custom made?)

Thank you for noticing. According to the Cambridge online dictionary, it is written with a hyphen, therefore, we added a hyphen and corrected the manuscript accordingly.

CRISPR-cas9 or CRISPR/Cas9? Keep it consistent throughout. CRISPR-cas9 is the latest consensus.

We corrected it to “CRISPR-Cas9” throughout the manuscript.

References are missing for several 'obvious statements' but please include them to reach a broader audience. For example the first 5 sentences of the introduction. Also, statements such as 'Root hairs are the primary entry point for rhizobial infection in most legumes.'

Thank you for the comment. To make it clearer, we also added reference #1, after the third sentence of the introduction, as well as we added an additional review as reference. This additional review was also cited as the source for the sentence “Root hairs are the primary...” (Please see page 2, revised manuscript)

Can you provide a percent value? Silencing of RIN4a and RIN4b resulted in significantly reduced nodule numbers on soybean transgenic roots in comparison to transgenic roots carrying the empty vector control. Also, this wording suggests it was a double K.D. but from the images, it appears they were individually silenced.

We appreciate the reviewer's comment. We observed a 50-70% reduction in the number of nodules. We adjusted the text according to the reviewer's remark. (See page 9, revised manuscript)

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